MR-Selection: A Meta-Reinforcement Learning Approach for Zero-Shot Hyperspectral Band Selection

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***Abstract*— Band selection is an effective method to deal with the difficulties in image transmission, storage, and processing caused by redundant and noisy bands in hyperspectral images (HSIs). Existing band selection methods usually need to learn a specific model for each HSI dataset, which ignores the inherent correlation and common knowledge among different band selection tasks. Meanwhile, these methods lead to a huge waste of computation. In this article, a novel zero-shot band selection method, called MR-Selection, is proposed for HSI classification. It formalizes zero-shot band selection as a metalearning problem, where advantage actor–critic algorithm-based reinforcement learning (A2C-RL) is designed to extract the metaknowledge in the band selection tasks of various seen hyperspectral datasets through a shared agent. To learn a consistent representation among different tasks, a dynamic structure-aware graph convolutional network is constructed to build a shared agent in A2C-RL. In A2C-RL, the state is tailored in a feasible way and easy to adapt to various tasks. Meanwhile, the reward is defined according to an efficient evaluation network, which can evaluate each state effectively without any fine-tuning. Furthermore, a two-stage optimization strategy is designed to coordinate optimization directions of a shared agent from different tasks effectively. Once the shared agent is optimized, it can be directly applied to unseen HSI band selection tasks without any available samples. Experimental results demonstrate the effectiveness and efficiency of the MR-Selection on the band selection of unseen HSI datasets.**

***Index Terms*— Band selection, graph convolution neural network (CNN), hyperspectral images (HSIs), metalearning, reinforcement learning (RL).**

I. Introduction

HYPERSPECTRAL imaging combines imaging technology with spectrum technology to obtain hyperspectral images (HSIs) with 2-D geometric spatial information and 1-D spectral information, which has the potential to identify fine ground objects [1]. HSIs have successful practical applications in many fields, such as biophysical parameter inversion [2], complex cross-scene HSI fine classification [3], [4], small target detection and classification [5], [6], and geological survey [7].

Due to the hyperspectral imaging mechanism, hundreds of spectral bands are acquired to form the HSIs. Substantial spectral bands lead to high computing costs and bring difficulties in the real-time processing of HSIs. Furthermore, HSIs also contain some noisy and redundant bands that will lead to the "Hughes phenomenon" [8]. Therefore, the dimensionality reduction of HSIs before further processing and analyzing has attracted more and more attention. At present, feature extraction and feature selection are the two mainstream methods in the field of dimensionality reduction [9]. In comparison with feature extraction, feature selection retains the original physical meaning of the data while reducing the data dimensionality.

The feature selection in HSIs is also known as band selection. Existing band selection methods for HSIs can be mainly divided into ranking-based methods, searching-based methods, clustering-based methods, sparsity-based methods, and embedding learning-based methods [10]. In the ranking-based method, the score of each band is calculated to measure the importance of each band by utilizing certain predefined criteria [11], [12]. Then, top-ranked bands are selected according to the sorted score. Minimum-variance principal component analysis (MVPCA) [12] is a representative ranking-based method that ranks the bands through a variance-based band power ratio.

In the searching-based methods, the criterion function is designed in advance to evaluate the candidate band subsets. In this way, the band selection problem can be solved by optimizing the designed criteria function. The feasible solution of the function is searched through the designed search strategy to form the appropriate band combination. The representative algorithms include the minimal-redundancy maximal-relevance (mRMR) framework [13] and the dynamic classifier selection (DCS) [14]. In mRMR, sequence forward or sequence backward searches are employed to explore the band combination according to the mRMR criterion. However, the band selection of mRMR has no correlation with the classification task, which limits the improvement of its classification performance. In [14], the band selection task of HSIs

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is modeled as the DCS process, where the superior classifiers are selected by the sequence forward selection strategy to obtain the appropriate band combination. mRMR combines multiple objective functions to form a single objective function. Compared with mRMR, the literature [15] designs the noninferior solution advantage matrix based on the swarm intelligence optimization method to solve the Pareto front of multiple objective functions. The search-based methods need to search for band combinations and evaluate the performance of each combination simultaneously, which needs a huge time cost than ranking-based methods.

The clustering-based methods divide the original spectral bands into several groups, and then, the ultimate band combination is constituted by selecting representative bands from each group. In [16], hierarchical clustering is performed to cluster the bands by minimizing the intracluster variance and maximizing the intercluster variance. Then, representative bands are acquired by reducing the data redundancy. Later, the tensor-based subspace clustering model [17] is proposed to jointly encode the multimode correlations of HSIs, so as to avoid destroying the internal spatial structure of HSIs.

The sparsity-based methods realize the 2band selection by setting some of the coefficients corresponding to unselected bands to zero, which can be regarded as an optimization problem with sparse constraints. This kind of method often reconstructs the original hyperspectral data approximately by the dictionary matrix and the coefficient matrix. In [18], a low-rank dictionary is learned in advance, and the coefficient matrix is learned and sparsely constrained. Then, an appropriate band combination is selected by clustering on the coefficient matrix. Different from [18], the dictionary matrix and sparse coefficient are added on an extra nonnegative constraint, which is learned simultaneously [19].

The embedding learning-based methods embed the band selection task into specific applications, such as HSI classification. RFE-SVM [20] is a typical embedded method. It sorts the weight value and eliminates the features during the training phase of SVM. Later, Kuo et al. [21] proposed a kernel-based band selection method on the basis of RFE-SVM for HSI classification. This method selects important bands by sorting the magnitude of SVM coefficients instead of the weight value. It promotes the band ranking according to the linear fitting degree of SVM and can better train the SVM classifier.

In the last decade, deep learning methods have developed vigorously in the field of band selection of HSIs. Among them, the convolution neural network (CNN) has become a research hotspot due to its powerful spatial extraction ability. According to the degree of utilization of label information, the CNN-based methods for band selection of HSIs mainly include three categories: supervised, unsupervised, and semisupervised.

Supervised methods implement the band selection under the guidance of label information [22], [23], [24], such as self-improving CNN (SICNN) [22] and distance density CNN (DDCNN) [23]. In SICNN, the candidate band subsets are searched by employing an improved particle swarm optimization. In order to effectively evaluate each candidate band subset, it is necessary to restart the training of 2-D CNN, which results in a high time cost. DDCNN [23] selects the band combinations based on the distance density. Then, a pretrained 1-D CNN with full-band data is directly used to evaluate the selected band subsets. Compared with SICNN, DDCNN greatly shortens the computation time to evaluate the band subset, but the pretrained network is not always effective in evaluating each candidate band subset. Unlike SICNN and DDCNN, the ternary weight CNN (TWCNN) [25] constructs a deepwise convolution layer and implements the band selection in an end-to-end manner. In TWCNN, a threshold-based ternary function is used to make the band weights sparse.

Since the label acquisition is labor intensive, a series of unsupervised methods are employed for the band selection task. BS-Nets and hybrid GCN (HGCN)-based band selection are representative methods [26], [27]. In BS-Nets, an end-to-end framework is proposed, where a band attention module is embedded in the reconstruction network to explicitly model the correlation between spectral bands, and a reconstruction network is used to reconstruct the original HSIs from the learned informative bands. Different from BS-Nets, HGCN simultaneously utilizes CNN and GCN to extract spatial–spectral features and refine the band correlation by the auxiliary spectrum reconstruction task. The improved versions of various reconstruction ways of BS-Nets are [28] and [29]. Semisupervised band selection methods utilize not only abundant unlabeled samples but also limited labeled samples as guidance. Sellami et al. [30] designed a method to select the bands according to the semisupervised low redundancy criterion and to extract the features from selected bands according to semisupervised 3-D CNN.

Recently, reinforcement learning (RL) gets more and more attention in the field of machine learning. RL has two obvious characteristics: trial-and-error and delayed reward mechanism [31]. RL is used to solve the sequential decision-making problem. Through interacting with the environment continuously, RL can select some actions and gets the most rewards as the best result until the completion of the episode. In addition, in order to solve complex tasks, deep learning and RL are usually combined, which is called deep RL (DRL). Due to the characteristics of DRL, it is very suitable for solving complex combinatorial optimization problems [32]. Coincidentally, the band selection task can be regarded as a sequence decision task of combinatorial optimization, which searches the optimal band combination by sequence forward in discrete space. In [33], the band selection task is framed with a tailored Markov decision process (MDP), and DRL is employed to solve the problem by using the information entropy as the reward. Feng et al. [34] proposed a new DRL modeling method for fixed and adaptive band selection of HSIs. Unlike [33], the literature [34] does not force the search space to be narrowed after each band is selected and defines a reward factor to avoid repeated selection of spectral bands.

Most of the band selection methods abovementioned have achieved good performance on specific HSI datasets. However, these methods have tailored a model for each specific dataset. When a new dataset is encountered, the model must be retrained to obtain a new set of parameters. In this case, these methods cannot achieve fast and effective inference

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on unseen datasets. The main reason for this problem is that the HSI datasets obtained by different HSI sensors are various in spatial resolution, spectral resolution, and spectral range, which makes it impossible for existing band selection methods to use the same model under fixed parameters for different HSI datasets. In fact, there is an inherent correlation among different band selection tasks. However, joint training in various tasks will lead to the explosion of the search space. Therefore, how to design a zero-shot hyperspectral band selection method that extracts the inherent correlation of different tasks and achieves superior search from a huge search space is a challenging topic.

Inspired by the idea of metalearning and RL, the zero-shot hyperspectral band selection problem is focused on in this article. Metalearning is used to learn the inherent correlation between band selection tasks from different seen HSI datasets. The band selection under each dataset is regarded as a subtask of the metalearning. Different subtasks are different sequential decision-making processes, which can be optimized by RL. Therefore, the zero-shot hyperspectral band selection is realized by dealing with the meta-RL. Specifically, advantage actor–critic algorithm-based RL (A2C-RL) is designed to constantly learn the metaknowledge from the continuous acquisition of past experience and efficiently search appropriate band combinations from the huge search space through a shared agent. In A2C-RL, the state representation is defined in a flexible way to adapt to different seen datasets. Meanwhile, an efficient evaluation network with random band sampling is designed to evaluate the reward from different datasets without fine-tuning. In addition, the dynamic structure-aware graph convolution network (GCN) is designed to construct a generalized policy network and critic network, namely, a shared agent. It can learn consistent feature representation from various seen datasets. Finally, the generalized policy and critic networks are alternately optimized to find appropriate band combinations by maximizing the reward. In addition, a two-stage optimization strategy is designed to alleviate the difficulty of optimizing different HSI seen tasks in diversity directions in A2C-RL. Once the shared agent is optimized, it can be directly applied to unseen HSI datasets to achieve zero-shot band selection of HSIs.

The main contributions of this article are listed as follows.

1) We apply the idea of meta-RL to the band selection of HSIs, making it possible to train on various seen datasets and inference on unseen datasets without any available samples for the hyperspectral band selection task. As far as we know, this is the first study to achieve zero-shot hyperspectral band selection.
2) The dynamic-aware GCN plays a role of the memory module in the metalearning. It eliminates the problem of inconsistent representation caused by different seen HSI datasets. Meanwhile, it continuously inherits the historical experience of previously learned tasks in the process of weight updating. Furthermore, a two-stage optimization strategy coordinates the diversity optimization objectives from different band selection tasks and enforces the agent to have outstanding generalization ability for unseen HSI band selection tasks.
3) In A2C-RL, the flexible state representation provides the potential to search across various seen HSI datasets. Moreover, the efficient evaluation network speeds up the searching process and guides MR-Selection to find satisfactory solution in a huge searching space.

II. Background

A. Metalearning

At present, most machine learning methods are aimed at a specific task, which results in considerable limitations. When the task changes, it is necessary for the model to be trained from scratch or fine-tuned. However, for applications such as real-time processing of remote sensing data in orbit, the acquisition of data labels is expensive, and the computing resources are very limited. Metalearning or "learning to learn" is another learning paradigm, which obtains distributed experience covering related tasks by observing multiple learning tasks, and then uses these experiences to improve model performance [35]. Metalearning draws on the idea of human learning. One of the key reasons why a human can learn quickly is that human can learn from experience. Metalearning forces the network to acquire the ability to learn through joint training of different tasks. When a similar new task appears, the acquired knowledge can be applied to quickly learn the new task. This way of "learning to learn" can have better performance in data demand and computational cost [36], and it is similar to the learning process of humans and animals [37], [38]. Recently, metalearning based on deep learning is mainly employed in zero-shot task [39], few-shot task [40], domain adaptive tasks [41], [42], and so on.

B. Graph Convolution Network

Traditional neural networks have achieved competitive results in processing data in the Euclidean space, but they have shown disappointing performance for data in the non-Euclidean space that is widely used in practical applications. However, graph neural networks (GNNs) show great potential in processing non-European data. Inspired by CNN, by introducing convolution operation into GNNs, GCN cleverly designed a method of extracting local features from graph data, which carries out end-to-end learning for node feature information and structure information. At present, GNNs are divided into two main schools: spectrum-based methods and space-based methods. As the most representative spectrum-based method, GCN has received more and more attention in a large number of research directions, such as image classification [43] and object detection [44]. In GCN, the graph is formulated as $G = \{V, E\}$, where V and E represent the node set and the edge set, respectively. For each node, it has its own feature $\mathbf{x}$. The features of N nodes are combined to form a feature matrix $\mathbf{X}_{N\times D}$, where D represents the dimension of the feature vector. Then, the spectral graph convolution based on the normalized Laplacian matrix of graph G is defined as follows:

$$\mathbf{x} *_G \mathbf{g}_\theta = \mathbf{U}\mathbf{g}_\theta\mathbf{U}^\top\mathbf{x} \tag{1}$$

where $\mathbf{x}$ represents the input data on the graph G, $*_G$ is the convolution operation, $\mathbf{g}_\theta = \{\text{diag}\}(\theta)$ is the filter, and

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$\mathbf{U}$ is the eigenvector matrix of normalized Laplacian matrix. Specifically, the basic idea [see (1)] is that $\mathbf{U}^\top\mathbf{x}$ is used to transform $\mathbf{x}$ to the spectral domain by making a graph Fourier transform, $\mathbf{g}_\theta\mathbf{U}^\top\mathbf{x}$ is to preform convolution operation on the transform result of $\mathbf{x}$, and $\mathbf{U}\mathbf{g}_\theta\mathbf{U}^\top\mathbf{x}$ is to transform convolution results to get the result of graph convolution by performing graph Fourier inverse transform. However, the number of parameters of the first-generation GCN is too large, resulting in high calculation costs and huge time waste. To address this issue, Kipf and Welling [45] designed an effective approach to improve GCN by further simplifying the assumptions on $\mathbf{g}_\theta$. In [45], GCN is represented as

$$\mathbf{x} *_G \mathbf{g}_\theta = \theta\left(\mathbf{I}_n + \mathbf{D}^{-\frac{1}{2}}\mathbf{A}\mathbf{D}^{\frac{1}{2}}\mathbf{x}\right) \tag{2}$$

where, due to the reduction of the number of parameters, the complexity of the model is reduced. After that, (2) is used widely.

C. Deep Reinforcement Learning

RL is used to solve the problem that how agents maximize reward or achieve specific goals through learning strategies in the process of interaction between agents and the environment, which is consistent with the goal of the MDP. Thus, the RL problem is usually modeled by MDP. An MDP is defined as a five-tuple $\langle S, A, T, R, \gamma\rangle$. Specifically, at each discrete time step t, the agent takes the state $s_t \in S$ obtained from the environment ε as input, and then, action $a_t \in A$ will be taken on the environment ε according to its policy π. Finally, the state is changed to s_{t+1} through transition function T, and the reward $r_t \in R$ is obtained in the current step.

RL optimization methods are mainly divided into policy-based and value-based. The representative method of policy-based optimization is REINFORCE [46]. In REINFORCE, the optimal policy is obtained by updating the following equation:

$$\nabla J(\theta) \approx \frac{1}{N}\sum_{i=1}^{N}\sum_{t=1}^{T}\left(\sum_{t'=t}^{T}\gamma^{t'-t}r_{i,t'} - b\left(s_{i,t}\right)\nabla_\theta \log\pi_\theta\left(a_{i,t}|s_{i,t}\right)\right) \tag{3}$$

where N is the number of episodes, T is the number of time steps in each episode, θ is the parameter of policy π, and $b(s_{i,t})$ is the baseline that represents expected accumulated reward at the state s_t in the ith episode. $\gamma \in (0, 1]$ is the discount factor.

The policy-based method is to directly parameterize the policy, which is simpler and easier to converge. However, the policy-based methods cause a huge time burden due to the requirement of completing the episode. Furthermore, the evaluation of policy is not sufficient, and the variance is large. Differently, the value-based methods are efficient in sample acquisition and have a small variance. The value-based methods do not learn a policy directly but learn a function to judge how good a state or a state–action pair is. Specifically, the value function $V^\pi(s)$ represents the expected accumulated reward at the state s according to the policy π. It is calculated as follows:

$$V_\pi(s) = \mathbb{E}\left[\sum_{t'=t}^{T}\gamma^{t'-t}r_{t'}\big|s_0 = s\right]. \tag{4}$$

The Q value function that represents the expected accumulated reward in the case of taking a according to the policy π in state s is utilized

$$Q_\pi(s, a) = \mathbb{E}\left[\sum_{t'=t}^{T}\gamma^{t'-t}r_{t'}\big|s_0 = s, \quad a_0 = a\right]. \tag{5}$$

The optimal Q^* value can be achieved by solving the following Bellman equation with the value iteration algorithm [47]:

$$Q^*(s, a) = \mathbb{E}\left[r + \gamma \max_{a'} Q^*\left(s', a'\right)\big|s, a\right]. \tag{6}$$

However, in the case of complex or continuous action space, the value-based method is always unsatisfactory. In order to alleviate the problem of the above two methods, the actor–critic (AC)-based method learns the optimal strategy by interactively training the policy network and the critic network [48]. Different from the policy-based methods, the AC-based methods do not directly use the feedback value of interaction with the environment but use the Q value function that represents the critic network to approximately represent the discount reward $\sum_{t'=t}^{T}\gamma^{t'=t}r_{i,t'}$ and calculate the policy gradient

$$\nabla J(\theta) \approx \frac{1}{N}\sum_{i=1}^{N}\sum_{t=1}^{T}\left(Q\left(s_{i,t}, a_{i,t}\right)\nabla_\theta \log\pi_\theta\left(a_{i,t}|s_{i,t}\right)\right). \tag{7}$$

At the same time, the critic network is updated according to the error of the Q value, and the loss function is

$$L_{\text{critic}}\left(\theta'\right) = \frac{1}{N} \times \sum_{i=1}^{N}\sum_{t=1}^{T}\left(r_{i,t} + \max Q\left(s_{i,t+1}, a_{i,t+1}, \theta'\right) - Q\left(s_{i,t}, a_{i,t}, \theta'\right)\right)^2 \tag{8}$$

where θ' is the parameter of the critic network.

A2C further improves the AC method by learning a better baseline for the AC method [49]. The A2C method estimates the baseline by the learnable approximate value function

$$b\left(s_{i,t}\right) = V^\pi\left(s_{i,t}\right). \tag{9}$$

Thus, the A2C calculates the policy gradient

$$\nabla J(\theta) \approx \frac{1}{N}\sum_{i=1}^{N}\sum_{t=1}^{T}\left(\left(Q\left(s_{i,t}, a_{i,t}\right) - V^\pi\left(s_{i,t}\right)\right)\nabla_\theta \log\pi_\theta\left(a_{i,t}|s_{i,t}\right)\right) \tag{10}$$

where $Q(s_{i,t}, a_{i,t}) - V^\pi(s_{i,t}) = A^\pi(s_{i,t}, a_{i,t})$ is called the advantage function, which is used to describe how much the action $a_{i,t}$ is better than expected.

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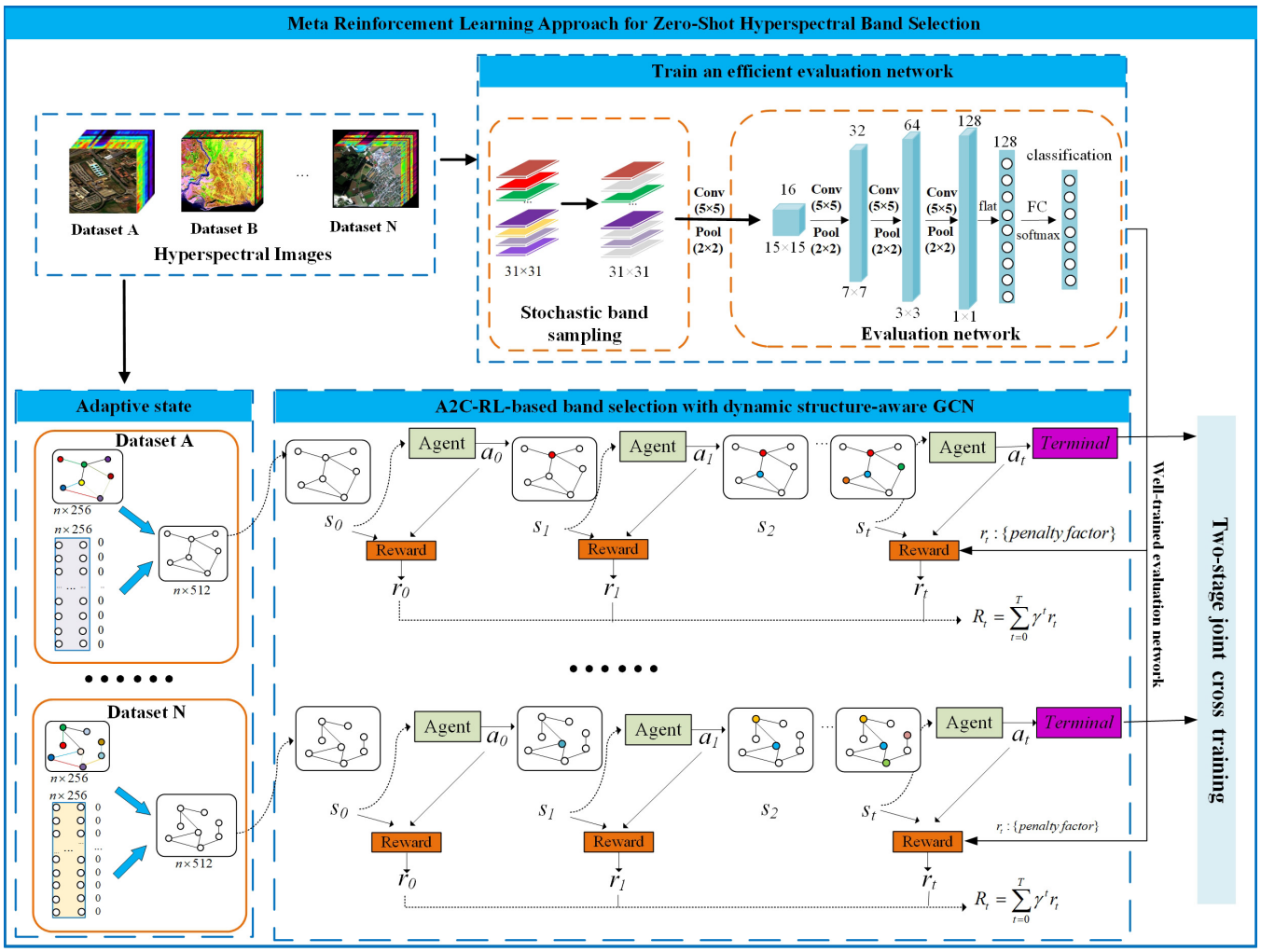


Fig. 1. Overview of the proposed MR-Selection method. It consists of two main stages: training an efficient evaluation network and training meta-RL agent for zero-shot band selection. In dynamic structure-aware GCN, colored nodes represent the corresponding selected bands, and uncolored nodes represent unselected ones.

III. Architecture of MR-Selection

Metalearning can formulate learning strategies to extract metaknowledge by exploring the inherent relevance of multiple tasks [50], [51]. Then, metaknowledge is used to guide the learning of unknown tasks, which makes the model have better generalization performance for new tasks. RL can constantly update the extracted information from historical experience in the process of iterative training of the agent and have a strong search ability in a huge search space [52], [53]. Inspired by metalearning and RL, a novel band selection method, MR-Selection, is designed for zero-shot hyperspectral band selection. The architecture of MR-Selection is shown in Fig. 1. As shown in Fig. 1, MR-Selection includes two main stages. In the first stage, an efficient evaluation network without fine-tuning is trained in advance to evaluate candidate band subsets. In the second stage, A2C-RL is implemented to extract the inherent correlation between different seen HSI datasets, namely, metaknowledge, where flexible state representations are designed to accommodate different datasets. Meanwhile, to alleviate the problem of inconsistent representation of different band selection tasks and aggregate historical information in the process of band selection, a dynamic structure-aware GCN is proposed to construct the shared agent with the policy network and the value network. Finally, a two-stage optimization strategy is used to jointly cross-train the shared agent in different band selection tasks. When the agent is optimized, a generalized policy network can be employed to guide the band selection process of unseen datasets.

A. Graph Representation

HSIs have abundant spectral bands from visible light to infrared spectrum through dense sampling, which makes it possible that there may be a correlation between adjacent bands or long-distance bands. The graph is an irregular data structure, which can flexibly model the correlation between the bands. Specifically, the band-based graph consists of nodes and edges. In MR-Selection, the training set from each HSI dataset for training RL agent is represented as $\mathbf{X}^r = \{\mathbf{x}_1^r, \mathbf{x}_2^r, \ldots, \mathbf{x}_N^r\}$. Each training sample is the spectral vector of length b, where b is the number of bands. Here, a fixed batch size from the training set $\mathbf{X}^r$ from each HSI dataset is chosen to construct the graph randomly. These samples are represented as $\overline{\mathbf{X}} = \{\overline{\mathbf{x}}_1, \overline{\mathbf{x}}_2, \ldots, \overline{\mathbf{x}}_{\overline{N}}\}$, where $\overline{N}$ represents the number of samples in fixed batch size. The structure of the band-based graph

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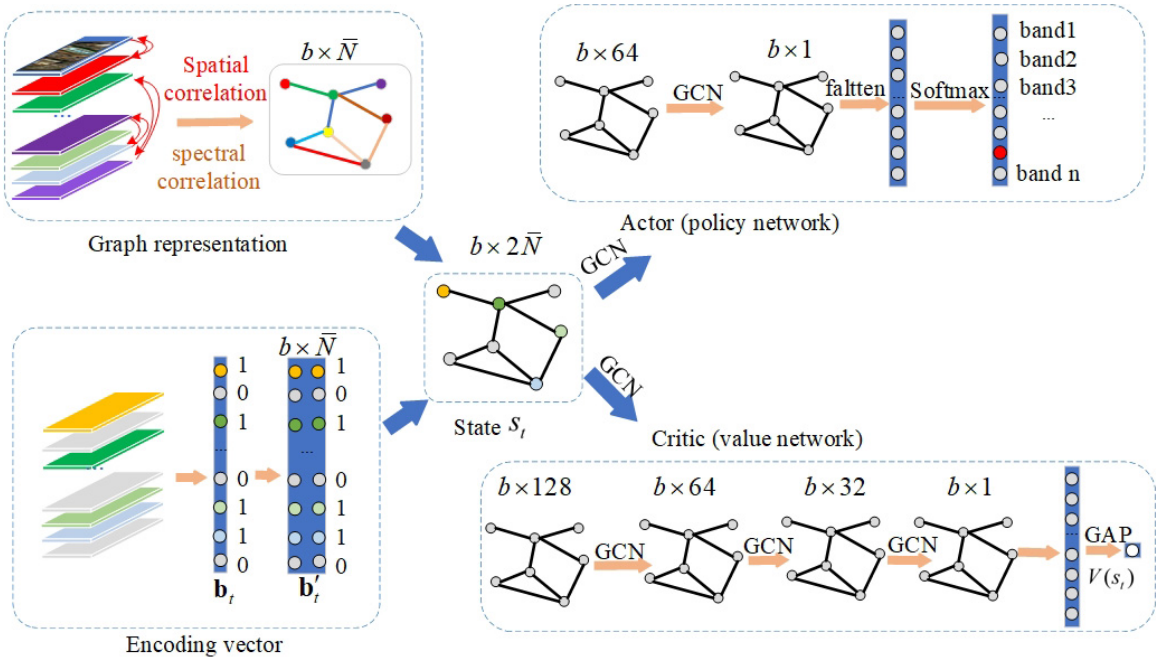


Fig. 2. Structure of the A2C-RL agent. It consists of a policy network and a value network.

is represented by $G = \{\mathbf{V}, \mathbf{A}, \mathbf{E}\}$. Assume that the graph $G = \{\mathbf{V}, \mathbf{A}, \mathbf{E}\}$ consists of a vertex set $\mathbf{V}$ and an edge set $\mathbf{E}$. $\mathbf{V} = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_b\}$ is a set of b vertices, where each vertex is a vector of length $\overline{N}$ and represents a band feature. $\mathbf{E}$ represents the edge relationship between spectral bands. $\mathbf{A} \in \mathbb{R}^{b\times b}$ is the corresponding adjacency matrix, and $\mathbf{A}_{ij}$ represents the correlation between the ith band and the jth band, which is calculated as follows:

$$\mathbf{A}_{ij} = \mathbf{B}_{ij} + \mathbf{S}_{ij} \tag{11}$$

$$\mathbf{B}_{ij} = \begin{cases} \exp(-|i-j|), & \text{if } i \neq j \\ 0, & \text{otherwise} \end{cases} \tag{12}$$

$$\mathbf{S}_{ij} = \begin{cases} \exp\left(-\sqrt{\frac{1}{\overline{N}}\sum_{k=1}^{\overline{N}}\left(\overline{\mathbf{x}}_k^i - \overline{\mathbf{x}}_k^j\right)}\right), & \text{if } i \neq j \\ 0, & \text{otherwise} \end{cases} \tag{13}$$

where $\mathbf{B}_{ij}$ indicates the spatial correlation between the ith and jth bands. When the ith and jth bands are closer in space, the value of $\mathbf{B}_{ij}$ is closer to 1. On the contrary, the value is closer to 0. In (13), $\mathbf{S}_{ij}$ represents the spectral correlation between the ith and ith bands. $\overline{\mathbf{x}}_k^i$ represents the spectral value of the ith band on the kth sample. In (12) and (13), a simplified Gaussian kernel function mapping ensures that the value of spatial correlation and spectral correlation is in the same range. Finally, $\mathbf{B}_{ij}$ and $\mathbf{S}_{ij}$ are summed to represent the correlation of the ith and jth bands.

B. Modeling the Band Selection Problem as MDP

To achieve zero-shot hyperspectral band selection, multiple seen datasets are trained to learn the inherent correlation between various band selection tasks. The sequence decision-making ability and strong search ability of RL make it very suitable for solving the task of band selection. The RL problem can be regarded as the MDP-based sequential decision-making problem [54]. In MR-Selection, the band selection process is transformed into the graph node selection process by employing graph structure to represent the correlation of spectral bands. Then, the graph node selection process is modeled as an MDP. A five-tuple $\langle S, A, T, R, \gamma \rangle$ is usually used to define an MDP. To enable MDP to be applied to the zero-shot hyperspectral band selection task, each element in the five-tuple is tailored as follows:

State: The state $s_t \in S$ in current time step t is defined as $G'_t = \{\mathbf{V} \oplus \mathbf{b'}_t, \mathbf{A}, \mathbf{E}\}$, which is modified on graph $G = \{\mathbf{V}, \mathbf{A}, \mathbf{E}\}$ and is described in Fig. 2. It can be seen from G'_t that the edge $\mathbf{E}$ and adjacency matrix $\mathbf{A}$ have not changed; only the vertex set has changed. As shown in Fig. 2, $\mathbf{b'}_t \in \mathbb{R}^{b\times\overline{N}}$ is an additional matrix, which is obtained by $\overline{N}$-dimensional columnwise expansion of the column vector $\mathbf{b}_t$ encoded by multihot encoding. The length of the column vector is equal to the number of bands. $\oplus$ represents the cascaded operation. In $\mathbf{b}_t$, 1 represents the selection of corresponding position nodes, and 0 corresponds to being unselected. For different numbers of spectral bands, the state can be adjusted flexibly, which is suitable for dynamically changing datasets in the joint cross-training process of multiple datasets.

Action: The action $a_t \in A$ is to select a node from the current state s_t at current time step t. It determines whether a certain band is selected or not. The probability of each band being selected is determined by the policy of the agent. After the agent takes a series of actions, the classification performance of the band combination is expected to be close to or even exceed the performance of the full bands.

State Transition Function: When the agent takes an action at the current time step t, the corresponding state s_t will be transferred to another state s_{t+1}. This process is realized by state transition function T. If a new node is selected, a new state $s_{t+1} = G'_t = \{\mathbf{V} \oplus \mathbf{b'}_{t+1}, \mathbf{A}, \mathbf{E}\}$ is obtained, where the

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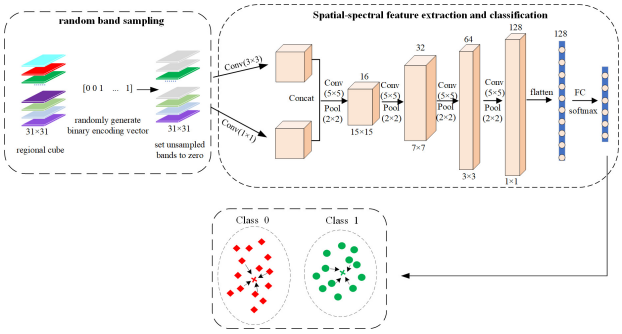


Fig. 3. Architecture of the evaluation network. It consists of a random band sampling part and spatial–spectral feature extraction and classification.

position corresponding to a new node in $\mathbf{b}'_{t+1}$ is set to 1. On the contrary, if a selected node is selected repetitively, the state keeps unchanged. The following formula represents the state transition function, as in (14), shown at the bottom of the next page.

Reward: The design of the reward function is very important, which determines the direction of the optimization of the agent. In MR-Selection, the classification performance of the selected node combination, namely, the band subset, is evaluated by an efficient evaluation network. The band subset needs to be evaluated until the number of selected bands reaches the preset value. In addition, the penalty factor is designed to punish the selection of duplicate bands, which avoids the agent from falling into a loop. The reward function for MR-Selection is defined as follows, as in (15), shown at the bottom of the next page, where *bandnum* represents the preset number of selected bands, which is a pregiven superparameter. Num$(\cdot)$ is used to count the number of elements of one in $\mathbf{b}_t$. $\text{loss}_{\text{full}} - \text{loss}_{\text{selected}}$ represents the evaluation score for the selected band subset in the current state by utilizing the evaluation network, $\text{loss}_{\text{full}}$ is the loss value of the training samples with all bands for evaluation through the evaluation network, and $\text{loss}_{\text{selected}}$ is the loss value of the training samples with selected bands for evaluation through the evaluation network, which will be described in detail in Section III-C.

C. Efficient Evaluation Network for Band Subset Evaluation

In some deep learning-based band selection approaches [22], [23], it is necessary to design a network to evaluate the band combinations generated during the search process. In DDCNN [23], a 1-D CNN is pretrained by the full-band HSI data, which may be inaccurate in the stage of evaluation. Unlike DDCNN, SICNN [22] needs to train 2-D CNN from scratch for each candidate band subset, which improves the evaluation accuracy to a certain extent but increases the computation cost. To alleviate the tradeoff of the evaluation network between accuracy and speed, we construct a special evaluation network. After the training is completed, the evaluation network can be implemented to evaluate any band subset directly and effectively.

The structure of the designed evaluation network is exhibited in Fig. 3. In MR-Selection, the training set from each HSI dataset for the evaluation network is represented as $\mathbf{X}^e = \{\mathbf{x}_1^e, \mathbf{x}_2^e, \ldots, \mathbf{x}_m^e\}$, in which every sample $\mathbf{x}_i^e \in \mathbb{R}^{w \times w \times b}$ is a cube centered on the current pixel and $w \times w$ is the spatial size of the cube. The labels of the training samples are represented by $\{y_1^e, y_2^e, \ldots, y_m^e\}$, $y_i^e \in \{1, 2, \ldots, K\}$, in which K represents the number of classes. In Fig. 3, the evaluation network mainly consists of two parts: random band sampling, and feature extraction and classification. For random band sampling, a binary vector with the same length as the number of spectral bands is randomly generated. Then, according to the binary encoding vector, the corresponding bands of training samples are set to zero. In this way, the evaluation network with random band sampling can explore more band combinations during the training. It tries to approximately include candidate band subsets that may appear during the search process and reduce the evaluation error.

For the feature extraction and classification, a simple CNN with spatial–spectral feature extraction and intraclass constraint is designed to improve the classification performance, where 1×1 and 3×3 convolution kernels are implemented to extract spectral and spatial features, respectively. Then, these two types of features are cascaded and input to the following convolutional layer. At the last full connection layer of the network, the softmax function is implemented to obtain the confidence that the training sample belongs to each class. In the stage of optimizing the network, the cross-entropy loss and the intraclass constraint are added as the final loss function to optimize the evaluation network. The intraclass constraint can force the network to narrow the distance from the same class of training samples, thereby achieving an excellent classification performance under the shortage of training samples. The loss of the evaluation network is calculated as

$$\text{loss} = L_a + L_c \tag{16}$$

where L_a is the intraclass constraint and L_c is the cross-entropy loss. The intraclass constraint is defined as follows:

$$L_a = \frac{1}{l} \sum_{i=1}^{l} \left\| \mathbf{f}_i^e - \mathbf{c}_j \right\| \tag{17}$$

$$\mathbf{c}_j = \frac{\sum_{y_i^e = j} \mathbf{f}_i^e}{S_j} \tag{18}$$

where l is the number of training samples in a fixed batch, $\mathbf{f}_i^e$ is the features of training sample $\mathbf{x}_i^e$ extracted from the last full connection layer of evaluate network, S_j is the number of training samples belonging to the jth class, and $\mathbf{c}_j$ denotes the class center of the jth class to which $\mathbf{f}_i^e$ belongs.

D. A2C-RL and Two-Stage Optimization Based on Dynamic Structure-Aware GCN for Zero-Shot Hyperspectral Band Selection

After the MDP is modeled for the node selection process and the evaluation network is well-trained, MR-Selection is optimized by using the A2C-RL to maximize the cumulative expected reward for the node selection problem. The A2C-RL method combines the policy-based method with the value-based method and has an obvious advantage in iteration stability and convergence speed. Since each node represents a band, the selected band subset is obtained when the node selection is completed. As shown in Fig. 2, the agent is constructed by a policy network and a value network in A2C-RL.

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To achieve the zero-shot band selection task, multiple seen HSI datasets are jointly cross-trained in both the policy and value networks. Since different datasets are constructed by various band-based graphs, the structure of the input graphs is dynamically changed during the training process. To adapt to the dynamically changing graph structure, dynamic structure-aware GCN is built for the policy and value networks. In dynamic structure-aware GCN, the fixed batch size of samples is selected to construct the graph for each dataset. Then, dynamic structure-aware GCN uses smart matrix multiplication to deal with different graph structures through the fixed-sized learned matrix. This learnable matrix can learn a unified correlation representation from different datasets. Meanwhile, it can be regarded as a memory module and continuously inherit past experience from different datasets.

After the policy and value networks are built by the dynamic structure-aware GCN, the policy network outputs the probability value of each action being selected, and the agent selects the action based on the probability value, which makes the agent to explore as many actions as possible. The value network outputs the single value, which represents the expected value of cumulative reward from the current state to the completion of the episode. Then, the value network assists in the optimization of the policy network during the alternate optimization of these two networks. In this way, the policy network can select the feasible band subset when the episode is completed.

Specifically, the forward propagation process in dynamic structure-aware GCN of the policy and value networks is formalized as follows:

$$\begin{aligned} \mathbf{H}_i^{l+1} &= o\left(\overline{\mathbf{A}}_i \mathbf{H}_i^l \mathbf{W}^l\right) \\ \overline{\mathbf{A}}_i &= \overline{\mathbf{D}}_i^{-\frac{1}{2}} (\mathbf{I}_i + \mathbf{A}_i) \overline{\mathbf{D}}_i^{\frac{1}{2}} \\ i &\in \{1, 2, \ldots, M\} \end{aligned} \tag{19}$$

where M is the number of training datasets. $\mathbf{H}_i^l$ and $\mathbf{H}_i^{l+1}$ represent the lth and $l+1$th layer features of the band-based graph With G_i corresponding to the ith dataset selected during the forward propagation process, $o(\cdot)$ represents the activation operation, I_i represents the identity matrix of G_i, $\mathbf{W}^l$ is fixed-sized learned matrix of the lth layer, and $\mathbf{A}_i$ is the adjacency matrix of G_i. $\overline{\mathbf{A}}_i$ is the adjacency matrix after regularization. $\overline{\mathbf{D}}_i$ refers to the degree matrix of G_i. The mth row and nth column element of $\overline{\mathbf{D}}_i$ is calculated by $\overline{\mathbf{D}}_i^{mm} = \sum_n (\mathbf{I}_i + \mathbf{A}_i)_{mn}$.

It is worth noting that the dimension of $\mathbf{W}^l$ is unrelated to the number of bands but only has a correlation with the fixed batch size of training samples. Specifically, $\overline{\mathbf{A}}_i \in \mathbb{R}^{b \times b}$, $\mathbf{H}_i \in \mathbb{R}^{b \times F_l}$, and $\mathbf{W}_i \in \mathbb{R}^{F_l \times F_{l+1}}$, where F_l and F_{l+1} represent the dimension of features in the lth and $l+1$th layers, and they can be designed artificially. Therefore, the matrix W of each layer can be kept a fixed size for different datasets through the input of fixed batch size of training samples. In this way, it can be shared between different datasets, which can learn the metaknowledge among different band selection tasks.

In order to alleviate the problem of optimizing in different directions for different band selection tasks, a two-stage optimization method is designed. Specifically, during the first stage optimization process, one gradient update is performed for each band selection task based on the parameters of the previous iteration. Then, one episode is sampled for each band selection task, and the corresponding agent loss is calculated with the parameters after one gradient update. At the second stage of optimization, the agent losses corresponding to all the band selection tasks are averaged, and the gradient is calculated; then, the second gradient update is performed based on the parameters of the previous iteration, that is, the metagradient update. The detailed procedure of MR-Selection is described in Algorithm 1.

IV. Experimental Result And Analysis

To verify the effectiveness of the MR-Selection, training and inferencing on the same dataset and the cross datasets are implemented. The specific organization of the experimental part is given as follows. First, six HSI datasets and experiment settings are described in Sections IV-A and IV-B. In Sections IV-C–IV-F, we discuss the classification results, time cost, comparison results with state-of-the-art unsupervised band selection methods, and the sensitivity to the number of the selected bands on the same dataset, respectively. In Sections IV-G–IV-I, we explain the classification results, sensitivity to the number of the selected bands, and selected spectral bands on the cross dataset. Finally, different penalty factors and ablation experiments are discussed in Sections IV-J and IV-K.

A. Data Description

In the experiment, six HSI datasets, namely, Indian Pines, Pavia University, University of Houston, Salinas Valley, KSC, and Washington D.C, are exploited to demonstrate the performance of the proposed MR-Selection. For each dataset except the Pavia University dataset, 5% labeled samples are randomly selected as the training set, and the rest samples are utilized

$$s_{t+1} = \begin{cases} \text{Terminal}, & \text{if } \text{Num}(\mathbf{b}_t) = \text{bandnum} \\ G'_t = \left\{\mathbf{V} \oplus \mathbf{b'}_{t+1}, \mathbf{A}, \mathbf{E}\right\}, & \text{if } a_t = \text{select a repetitive node} \\ G'_{t+1} = \left\{\mathbf{V} \oplus \mathbf{b'}_{t+1}, \mathbf{A}, \mathbf{E}\right\}, & \text{if } a_t = \text{select a new node} \end{cases} \tag{14}$$

$$r_t = \begin{cases} \text{loss}_{s_{\text{full}}} - \text{loss}_{\text{selected}}, & \text{if } \text{Num}(\mathbf{b}_t) = \text{bandnum} \\ r_t = \alpha, & \text{if } a_t = \text{select a repetitive node} \\ r_t = 0, & \text{if } a_t = \text{select a new node} \end{cases} \tag{15}$$

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Algorithm 1 MR-Selection

Input : training dataset $X^{tr} = \{X_1^{tr}, X_2^{tr}, \ldots, X_n^{tr}\}$; training label set $Y = \{Y_1, Y_2 \cdots, Y_n\}$; the number of selected bands k; the first stage optimization learning rate α,the meta learning rate β;

Output: the selected bands and the predicted labels of test samples in the test dataset.

Training of MR-Selection:
Training an evaluation network for every HSI dataset;
Initialization the parameter θ_a of the policy network, θ_v of the value network;
for *every epoch* **do**
  for *each band selection task* T_i **do**
    Set the gradient of θ_a and θ_v of T_i to 0: $d_{T_i}\theta_a = 0$, $d_{T_i}\theta_v = 0$
    Construct a graph G from the selected HSI dataset by (11)-(13)
    Initialize the state s_t with the constructed graph G
    while *the number of selected nodes does not reach* k **do**
      Perform a_t according to the policy $\pi_{\theta_a}(a_t|s_t)$
      Receive the reward r_t by (15) and generate the new state s_{t+1} by (14)
    end
    $R = 0$
    for $j \in \{k, \ldots, 1\}$ **do**
      $R \leftarrow r_j + \gamma R$
      Accumulate gradients of θ_a for T_i:
      $d_{T_i}\theta_a = d_{T_i}\theta_a + \nabla_{\theta_a} \log \pi(a_j|s_j;\theta_a)(R - V(s_j|\theta_v))$
      Accumulate gradients of θ_v for T_i:
      $d_{T_i}\theta_v = d_{T_i}\theta_v + \frac{tial(R - V(s_j;\theta_v))^2}{tial\theta_v}$
    end
    Calculate parameters:$\theta_a' = \theta_a - \alpha d_{T_i}\theta_a$, $\theta_v' = \theta_v - \alpha d_{T_i}\theta_v$
    Sample a trajectory of current band selection task using θ_a', θ_v'
    Calculate $d_{T_i}\theta_a', d_{T_i}\theta_v'$ according to the calculation process of $d_{T_i}\theta_a, d_{T_i}\theta_v$
  end
  calculate $d_{\theta_a} = \frac{1}{n}\sum_{i=1}^{n} d_{T_i}\theta_a'$, $d_{\theta_v} = \frac{1}{n}\sum_{i=1}^{n} d_{T_i}\theta_v'$
  Update policy and value network:$\theta_a = \theta_a - \beta d\theta_a$, $\theta_v = \theta_v - \beta d\theta_v$
end
Test of MR-Selection:
Output the selected bands and labels of test samples by the well-trained RL agent

as the test set. For Pavia University, 3% labeled samples are randomly selected as the training set to train all the algorithms, and the rest samples are utilized as the test set. The details of the six datasets are introduced as follows.

1) Indian Pines: This dataset was acquired by employing the Airborne Visual Infrared Imaging Spectrometer (AVIRIS) in Indiana, USA, in 1992, which consists of 145 × 145 pixels and 200 spectral bands. Sixteen classes are included in this dataset, and the number of samples of each class is exhibited in Table I. Moreover, to have a more intuitive display, the false-color image formed by extracting 50, 27, and 17 bands is displayed in Fig. 4(a).

TABLE I
16 CLASSES AND SAMPLE NUMBER OF EVERY CLASS ON THE INDIAN PINES DATASET

#	Class	Samples
1	Alfalfa	46
2	Corn-notill	1428
3	Corn-mintill	830
4	Corn	237
5	Grass-pasture	483
6	Grass-trees	730
7	Grass-pasture-mowed	28
8	Hay-windrowed	478
9	Oats	20
10	Soybean-notill	972
11	Soybean-mintill	2455
12	Soybean-clean	593
13	Wheat	205
14	Woods	1265
15	Buildings-Grass-Trees-Drives	386
16	Stone-Steel-Towers	93
	Total	10249

TABLE II
NINE CLASSES AND SAMPLE NUMBER OF EVERY CLASS ON THE PAVIA UNIVERSITY DATASET

#	Class	Samples
1	Asphalt	6631
2	Meadows	18649
3	Gravel	2099
4	Trees	3064
5	Painted metal sheets	1345
6	Bare Soil	5029
7	Bitumen	1330
8	Self-Blocking Bricks	3682
9	Shadows	947
	Total	42776

2) Pavia University: This dataset was obtained by using the German Reflective Optics Spectrographic Imaging System (ROSIS-03) in Italy in 2003. ROSIS-03 continuously images 115 bands in the 0.43–0.86-μm wavelength range. Removing 12 bands affected by noise, the remaining 103 bands are used. Table II lists the nine classes and the number of samples of each class in this dataset. The false-color image formed by extracting 53, 31, and eight bands is shown in Fig. 4 (b).

3) University of Houston: This dataset was acquired by the ITRES CASI-1500 sensor and provided by the 2013 IEEE Geoscience and Remote Sensing Society (GRSS) Data Fusion Competition. This dataset consists of 349 × 1905 pixels and 144 spectral bands with a spectral range from 364 to 1046 nm. This scene contains a total of 15 classes. In addition, the number of each class is listed in Table III. The false-color image formed by extracting 28, 45, and 65 bands is shown in Fig. 4(c).

4) Salinas Valley: This dataset was taken by an AVIRIS sensor in the Salinas Valley, California. The spatial size of the image is 512 × 217. The original data has 224 spectral bands. After removing the bands with severe water vapor absorption,

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TABLE III
FIFTEEN CLASSES AND SAMPLE NUMBER OF EVERY CLASS ON THE UNIVERSITY OF HOUSTON DATASET

#	Class	Samples
1	gras_healthy	1251
2	grass_stressed	1254
3	grass_synthetic	697
4	tree	1244
5	soil	1242
6	water	325
7	residential	1268
8	commercial	1244
9	road	1252
10	highway	1227
11	railway	1235
12	parking_lot1	1233
13	parking_lot2	469
14	tennis_court	428
15	running_track	660
	Total	15029

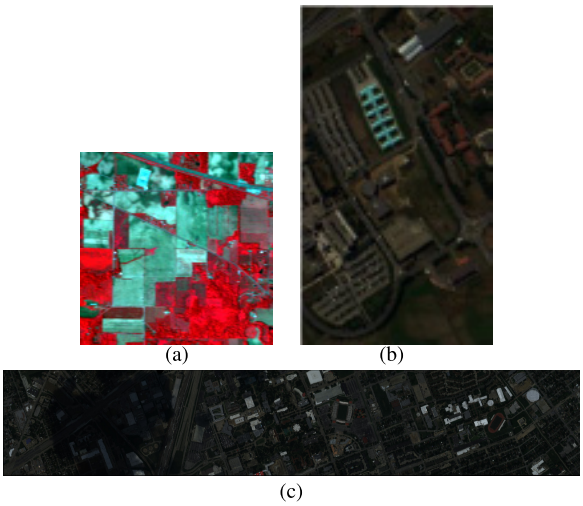


Fig. 4. False-color composite images of (a) Indian Pines, (b) Pavia University, and (c) University of Houston.

there are 204 spectral bands. This dataset contains 16 crop classes.

5) Washington D.C: This dataset is acquired by the Hydice sensor and contains a total of 191 bands ranging from 0.4- to 2.4-μm visible light and near-infrared. The spatial size of the image is 750 $\times$ 307. This dataset contains nine classes.

6) KSC: This dataset was obtained by the AVIRIS sensor, which contains 224 spectral bands. After removing the bands with severe water vapor absorption, the remaining 176 bands are used. The spatial size of the image is 512 $\times$ 616, and there are 13 classes in total.

B. Experimental Settings

To testify to the efficiency and effectiveness of the MR-Selection algorithm, we choose seven representative band selection algorithms for comparison on the same dataset and unseen dataset, including mRMR [13], DDCNN [23], BS-Nets [26], TWCNN [25], SICNN [22], DRLBS [33], and DCS [14]. Experiments on the same dataset mean that training and testing use the same dataset. MR Selection without two-stage optimization is used to implement experiments on the same dataset. Experiments on the unseen dataset mean that training and testing use different datasets. Moreover, in order to compare with the classification results of the full band, SVM with the radial basis kernel function (RBF-SVM) is added to perform the classification task. The proposed MR-Selection algorithm is implemented in python language with Pytorch framework. In addition, all experiments are carried out in a computer equipped with an Intel i7-7820X CPU and Nvidia RTX 2080Ti GPU.

For RBF-SVM, a one-against-all strategy is implemented to achieve the classification task, and the grid search is implemented to find the optimal penalty and gamma parameters. For mRMR, the forward selection tries to band subset according to the minimal-redundancy-maximal-relevance criterion, and SVM is employed as a classifier to evaluate the performance of the selected bands. For DDCNN and TWCNN, all the parameters are set to be the same as those in [23] and [25], respectively. In DCS, the number of neighbors in the K-nearest neighbor is set to 10, and SVM is employed as the classifier. In SICNN, all the parameters in fractional order Darwinian particle swarm optimization are set according to the suggestion in [22], but the spatial size of the input region cube in 2-D-CNN is set to 27 $\times$ 27. In BS-Nets, the learning rate, the number of epochs, and the regularization coefficient are set to 0.001, 500, and 0.02, respectively. For DRLBS, it treats the band selection task as the MDP of an agent that uses the RL approach to solve, in which information entropy and correlation coefficient are measured as a reward scheme. In DRLBS, all parameters are set to default values in [33]. For MR-Selection, the training samples in each training dataset are equally divided into two parts $\{\mathbf{X}^r, \mathbf{X}^e\}$, where $\mathbf{X}^r$ is the training set for training of the agent and 256 fixed batch size of $\mathbf{X}^r$ is chosen randomly to construct the graph. $\mathbf{X}^e$ is the training set for the evaluation network. The penalty factor α is set to -0.0001 by the trial-and-error procedure. The first stage optimization learning rate and the metalearning rate are set to 0.01, and the number of epochs of the agent is set to 5000, respectively. In the process of training evaluation network, the spatial size of the input region cube is 31 $\times$ 31. The learning rate, batch size, and epoch are set to 0.0005, 128, and 5000, respectively.

C. Classification Results on the Same Dataset

In this part, all the compared methods train and inference on the same dataset. Three indices, the overall accuracy (OA), the average accuracy (AA), and the Kappa coefficient (Kappa), are employed to quantify the classification performance.

1) Indian Pines: For all the band selection algorithms, 60 bands among 200 spectral bands are selected in this dataset. The average and standard deviation of the classification results after running 30 times independently are recorded in the table. In the table, the optimal results of each class, OA, AA, and Kappa are highlighted in the gray region.

As shown in Table IV, the classification performance of BS-Nets is unsatisfying among all the algorithms. Although BS-Nets preserves the most important bands by reconstructing the data, the label information is not used in the process of

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TABLE IV

CLASSIFICATION RESULTS OF RBF-SVM, MRMR, SICNN, DDCNN, DRLBS, DCS, BS-NETS, TWCNN, AND MR-SELECTION ON THE INDIAN PINES DATASET

Class	RBF-SVM	mRMR	SICNN	DDCNN	DRLBS	DCS	BS-Nets	TWCNN	MR-Selection
1	55.8 ± 14.1	35.9 ± 23.0	83.4 ± 7.0	29.5 ± 9.0	43.4 ± 18.6	46.2 ± 15.4	16.2 ± 11.9	68.2 ± 21.4	96.1 ± 2.0
2	74.6 ± 1.7	66.7 ± 5.5	81.9 ± 4.9	77.4 ± 0.4	72.6 ± 5.3	68.5 ± 3.9	67.5 ± 2.3	89.1 ± 0.5	97.0 ± 0.6
3	63.4 ± 4.4	50.2 ± 5.4	82.8 ± 3.2	61.3 ± 5.0	63.1 ± 1.8	53.3 ± 1.6	50.9 ± 4.4	92.9 ± 1.7	95.3 ± 0.9
4	42.1 ± 5.1	50.4 ± 8.3	85.4 ± 2.3	42.0 ± 6.7	45.9 ± 15.8	53.7 ± 2.5	42.6 ± 3.0	90.9 ± 2.5	99.1 ± 0.7
5	88.7 ± 2.7	83.3 ± 4.1	79.9 ± 8.4	83.2 ± 1.9	84.7 ± 3.7	84.3 ± 2.6	80.8 ± 5.3	86.1 ± 5.9	92.0 ± 1.5
6	95.6 ± 2.2	89.6 ± 2.3	96.3 ± 2.0	89.6 ± 1.5	92.4 ± 2.2	90.8 ± 4.5	93.0 ± 1.8	95.5 ± 3.0	94.7 ± 1.4
7	70.8 ± 11.1	40.8 ± 32.0	82.4 ± 3.9	89.6 ± 2.1	62.7 ± 32.6	45.8 ± 33.3	30.6 ± 22.1	93.8 ± 8.7	94.4 ± 7.8
8	97.4 ± 1.0	94.9 ± 3.5	86.0 ± 9.8	94.5 ± 0.6	92.5 ± 3.7	94.5 ± 3.8	95.6 ± 1.3	100.0 ± 0.0	99.9 ± 0.2
9	24.2 ± 13.7	11.8 ± 11.8	42.2 ± 9.8	38.2 ± 8.8	17.6 ± 28.0	26.5 ± 8.8	11.8 ± 0.0	87.7 ± 2.5	55.6 ± 13.6
10	72.1 ± 3.3	68.5 ± 3.9	78.4 ± 3.7	72.3 ± 3.3	68.4 ± 4.3	64.0 ± 3.8	60.0 ± 6.6	89.5 ± 2.7	93.5 ± 1.0
11	79.1 ± 2.3	75.2 ± 0.8	94.1 ± 3.4	83.5 ± 0.3	76.1 ± 2.4	74.9 ± 0.9	67.6 ± 1.3	96.4 ± 2.8	98.1 ± 0.2
12	60.9 ± 3.8	46.6 ± 4.3	78.3 ± 6.8	62.8 ± 0.7	63.5 ± 7.3	58.3 ± 2.2	35.9 ± 4.4	76.3 ± 4.0	98.6 ± 0.3
13	97.2 ± 1.5	82.3 ± 6.7	82.3 ± 8.0	95.1 ± 0.3	94.9 ± 4.3	92.5 ± 2.3	91.2 ± 4.7	98.6 ± 0.9	98.5 ± 0.4
14	91.4 ± 2.7	90.5 ± 2.7	97.7 ± 1.4	94.8 ± 2.1	91.8 ± 3.3	84.7 ± 2.9	89.8 ± 4.1	97.2 ± 1.4	98.6 ± 0.7
15	43.3 ± 6.3	34.5 ± 14.1	83.6 ± 3.3	40.5 ± 0.3	38.5 ± 6.7	58.7 ± 6.2	40.3 ± 3.0	87.5 ± 6.3	99.0 ± 0.7
16	89.2 ± 5.0	88.1 ± 3.3	88.3 ± 6.3	86.1 ± 7.6	70.9 ± 34.5	94.3 ± 0.6	84.0 ± 3.3	91.4 ± 11.4	58.3 ± 1.5
OA(%)	77.5 ± 0.4	71.7 ± 0.6	87.6 ± 1.0	78.3 ± 0.3	75.2 ± 1.2	73.1 ± 0.1	68.7 ± 0.9	92.4 ± 1.0	96.5 ± 0.1
AA(%)	71.6 ± 1.6	63.1 ± 3.4	82.7 ± 2.6	71.3 ± 0.2	67.4 ± 1.6	68.2 ± 0.5	59.9 ± 1.8	90.1 ± 2.4	91.8 ± 1.3
Kappa(%)	74.3 ± 0.5	67.6 ± 0.7	85.9 ± 1.1	75.1 ± 0.3	71.7 ± 1.4	69.2 ± 0.0	64.3 ± 1.0	91.3 ± 1.1	96.0 ± 0.1

TABLE V

CLASSIFICATION RESULTS OF RBF-SVM, MRMR, SICNN, DDCNN, DRLBS, DCS, BS-NETS, TWCNN, AND MR-SELECTION ON THE PAVIA UNIVERSITY DATASET

Class	RBF-SVM	mRMR	SICNN	DDCNN	DRLBS	DCS	BS-Nets	TWCNN	MR-Selection
1	88.5 ± 1.0	85.5 ± 1.9	93.0 ± 1.9	88.3 ± 1.2	90.7 ± 1.1	84.6 ± 3.3	84.6 ± 5.8	96.2 ± 0.9	99.9 ± 0.0
2	97.5 ± 0.2	95.1 ± 2.1	99.0 ± 0.4	95.6 ± 1.7	95.6 ± 0.8	95.1 ± 1.3	97.2 ± 0.6	99.6 ± 0.2	100.0 ± 0.0
3	65.5 ± 3.1	31.0 ± 8.1	96.1 ± 2.5	43.5 ± 14.8	72.0 ± 1.0	49.1 ± 3.7	27.5 ± 8.4	65.2 ± 8.9	98.9 ± 0.4
4	90.0 ± 1.6	76.8 ± 5.4	94.8 ± 2.6	80.9 ± 3.6	88.9 ± 1.9	82.4 ± 1.3	78.4 ± 1.8	97.6 ± 0.6	97.9 ± 0.2
5	99.3 ± 0.1	99.0 ± 0.4	99.2 ± 0.2	99.3 ± 0.2	99.0 ± 0.4	99.1 ± 0.2	99.2 ± 0.2	100.0 ± 0.0	100.0 ± 0.0
6	61.1 ± 1.1	40.3 ± 7.9	87.2 ± 1.8	57.3 ± 7.4	67.9 ± 3.0	53.2 ± 3.9	31.5 ± 8.9	96.0 ± 1.3	100.0 ± 0.0
7	76.6 ± 4.5	55.2 ± 6.0	78.1 ± 1.7	59.7 ± 8.0	73.6 ± 3.0	69.8 ± 5.5	45.4 ± 12.7	84.8 ± 3.3	95.8 ± 1.9
8	85.5 ± 2.6	81.8 ± 1.7	96.6 ± 2.0	83.1 ± 6.8	83.1 ± 1.7	82.1 ± 1.5	80.4 ± 2.7	94.4 ± 0.3	99.1 ± 0.3
9	95.9 ± 2.1	99.8 ± 0.2	93.6 ± 4.1	99.8 ± 0.2	99.6 ± 0.1	99.8 ± 0.1	99.8 ± 0.1	98.6 ± 1.0	97.4 ± 0.7
OA(%)	88.0 ± 0.5	80.5 ± 0.5	95.3 ± 0.9	84.4 ± 0.5	88.4 ± 0.4	83.7 ± 0.5	79.8 ± 0.9	95.9 ± 0.8	99.5 ± 0.1
AA(%)	84.4 ± 0.6	73.8 ± 0.3	93.1 ± 0.7	78.6 ± 1.5	85.6 ± 0.6	79.5 ± 0.5	71.6 ± 2.0	92.5 ± 1.5	98.8 ± 0.2
Kappa(%)	83.9 ± 0.6	73.5 ± 0.6	93.7 ± 0.7	78.9 ± 0.9	84.4 ± 0.6	78.0 ± 0.7	72.2 ± 1.4	95.5 ± 1.0	99.3 ± 0.1

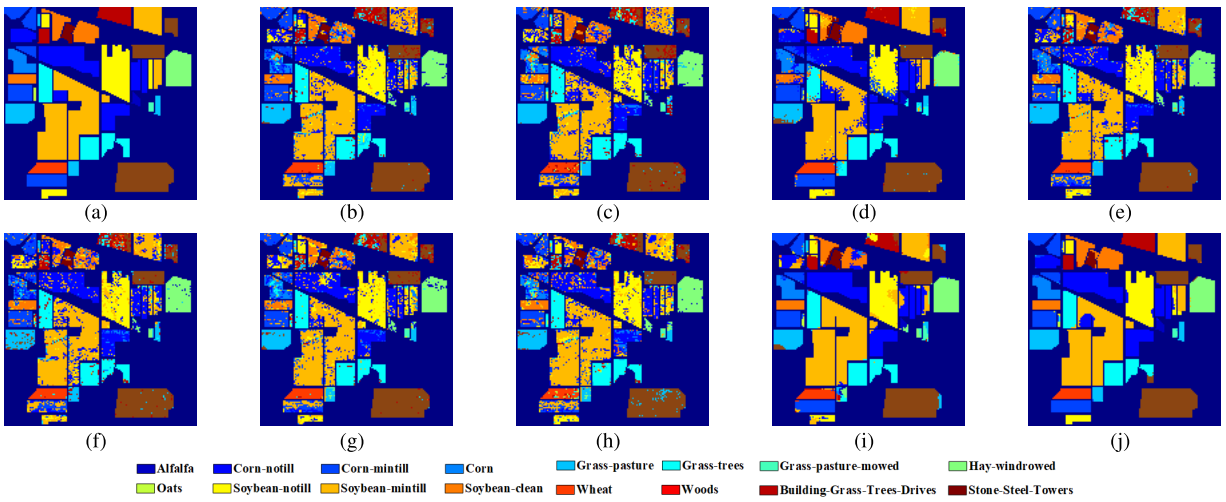


Fig. 5. (a) Ground truth and (b)–(j) visual classification result map on the Indian Pines dataset by RBF-SVM, mRMR, SICNN, DDCNN, DRLBS, DCS, BS-Nets, TWCNN, and MR-Selection.

band selection. The mRMR achieves slightly better performance than BS-Nets. However, the band selection process and the classification task have no correlation in mRMR. DCS outperforms mRMR because it transforms the band selection into the selection of classifiers so that the classification task can guide the band selection task directly. DRLBS performs better than DCS due to effective optimization in a trial-and-error manner. Thanks to the powerful feature representation capabilities of deep learning, the CNN-based algorithms, DDCNN, SICNN, TWCNN, and MR-Selection, show better

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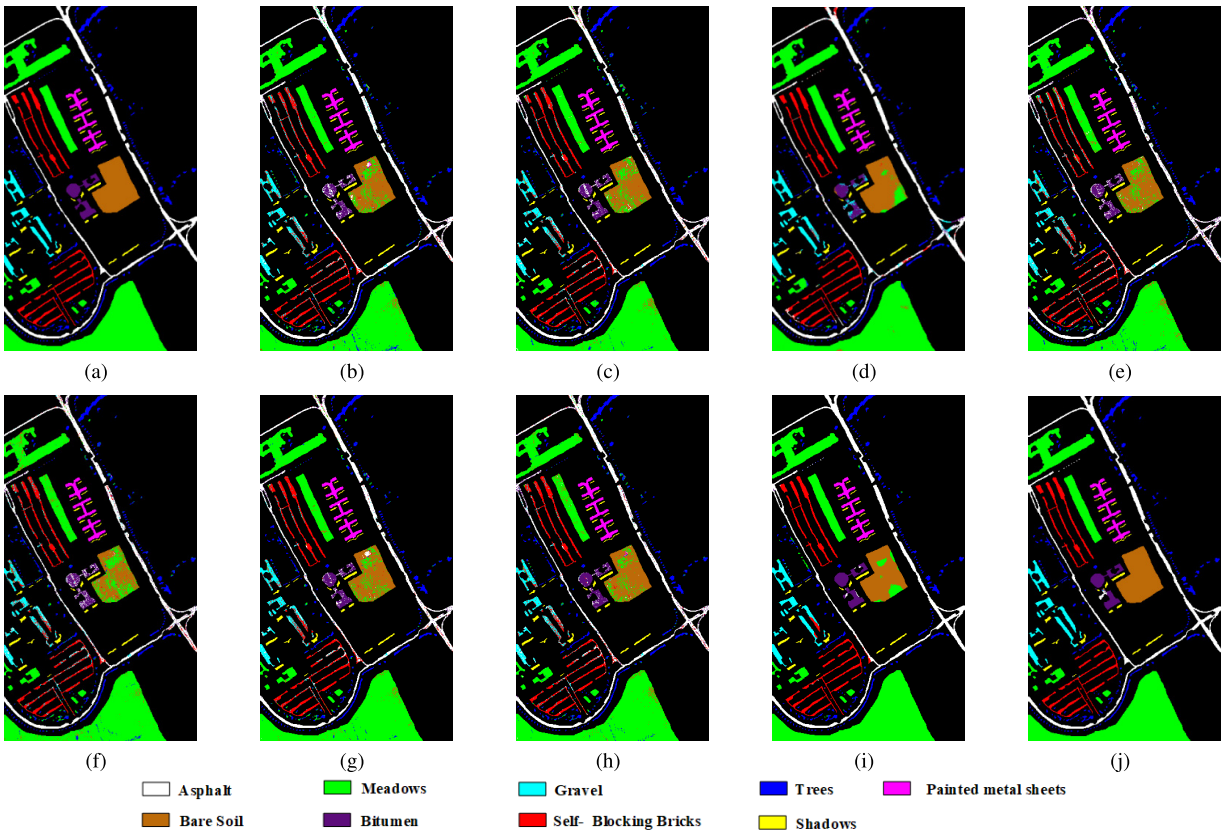


Fig. 6. (a) Ground truth and (b)–(j) visual classification maps on the Pavia University dataset by RBF-SVM, mRMR, SICNN, DDCNN, DRLBS, DCS, BS-Nets, TWCNN, and MR-Selection.

performance than other comparison algorithms. SICNN is superior to DDCNN by training 2-D CNN to evaluate each band subset in the band selection process. TWCNN is better than DDCNN and SICNN. Among the four deep learning-based algorithms, MR-Selection improves the OA, AA, and Kappa indexes by at least 4.1%, 1.7%, and 4.7% through efficient band selection evaluation and powerful search capability of A2C-RL. To obtain a more intuitive display, the visual classification result maps of all the nine algorithms and ground truth are exhibited in Fig. 5. As shown in Fig. 5(b), (c), and (e)–(h), the visual classification result maps contain substantial misclassified samples and noisy points. The main reason is that RBF-SVM, BS-Nets, DRLBS, mRMR, DCS, and DDCNN input the samples in the form of pixels in the stages of band selection and classification. SICNN, TWCNN, and MR-Selection use the regional cube by incorporating spatial information. Thus, these methods have better regional consistency in the visualized classification maps. However, there are still obvious misclassified samples at the adjacent borders of the regions on SICNN and TWCNN, such as soybean-notill and grass-pasture classes. Compared to SICNN and TWCNN, MR-Selection can obtain a better boundary location on corn-notill and soybean-notill classes and improve regional consistency.

2) Pavia University: For all the band selection algorithms, 30 bands among 103 spectral bands are selected in this dataset. Table V shows the quantitative classification results of eight comparison algorithms and MR-Selection. As shown in Table V, MR-Selection has achieved excellent classification performance in most categories. In particular, MR-Selection has obtained a significant improvement compared to other comparison algorithms in the AA, OA, and Kappa indexes. Compared to TWCNN, MR-Selection achieves at least 3.6%, 6.3%, and 3.8% improvements in terms of OA, AA, and Kappa indexes. In addition, it is found that most of the comparison algorithms have shown poor distinguishing ability on gravel, bare soil, and bitumen classes. However, MR-Selection has an increase of at least 2.8%, 11%, and 2.5% improvements due to the powerful band evaluation and searching capabilities.

The visual classification result maps of all the nine algorithms and the ground truth are displayed in Fig. 6. As shown in Fig. 6(b)–(i), most samples in the bare soil class are incorrectly classified as the meadows class. On the contrary, MR-Selection can achieve a satisfactory classification performance in all the classes. Especially, compared to RBF-SVM, mRMR, DDCNN, DRLBS, DCS, and BS-Nets, MR-Selection has almost no noisy scattered points and achieves a smoother effect, which improves the regional consistency.

3) University of Houston: For all the band selection algorithms, 40 bands among 176 spectral bands are selected in this dataset. Table VI exhibits the quantitative classification results of eight comparison algorithms and MR-Selection. As shown in Table VI, MR-Selection has obtained the best classification performance in most classes. In addition, in terms of OA, AA, and Kappa indexes, MR-Selection has increased by at least 2.5%, 2.6%, and 2.7%, respectively.

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TABLE VI

CLASSIFICATION RESULTS OF RBF-SVM, MRMR, SICNN, DDCNN, DRLBS, DCS, BS-NETS, TWCNN, AND MR-SELECTION ON THE HOUSTON DATASET

Class	RBF-SVM	mRMR	SICNN	DDCNN	DRLBS	DCS	BS-Nets	TWCNN	MR-Selection
1	93.1 ± 1.7	95.4 ± 1.3	97.6 ± 0.7	94.4 ± 2.3	94.6 ± 2.8	97.1 ± 0.6	91.7 ± 7.0	93.1 ± 3.0	97.0 ± 0.6
2	96.7 ± 0.4	89.5 ± 5.0	98.5 ± 0.5	96.4 ± 2.6	97.8 ± 1.7	95.9 ± 2.1	89.7 ± 4.5	97.8 ± 1.3	96.7 ± 0.8
3	97.4 ± 0.3	87.4 ± 4.0	98.5 ± 0.4	98.3 ± 1.4	99.2 ± 0.5	98.3 ± 1.8	88.5 ± 0.3	99.8 ± 0.2	97.7 ± 0.3
4	92.6 ± 2.1	90.9 ± 0.3	98.7 ± 0.1	94.6 ± 1.5	97.2 ± 2.2	95.8 ± 2.9	95.8 ± 2.9	92.5 ± 2.3	97.8 ± 0.9
5	97.2 ± 0.2	98.4 ± 0.5	96.5 ± 0.5	97.3 ± 1.2	98.2 ± 0.9	98.6 ± 0.3	96.9 ± 0.0	99.3 ± 0.4	98.5 ± 1.2
6	94.2 ± 0.4	94.6 ± 2.9	93.6 ± 1.4	96.0 ± 0.7	93.6 ± 5.6	89.7 ± 4.6	87.5 ± 7.5	88.1 ± 1.2	92.2 ± 0.7
7	94.6 ± 0.6	73.0 ± 3.1	91.4 ± 0.9	89.8 ± 0.3	87.7 ± 2.7	89.5 ± 2.4	78.2 ± 1.6	93.3 ± 1.5	96.0 ± 0.8
8	76.2 ± 3.8	75.0 ± 3.6	83.8 ± 2.9	78.4 ± 2.0	82.6 ± 2.4	80.3 ± 3.9	77.8 ± 1.5	88.5 ± 6.1	87.2 ± 0.9
9	89.5 ± 1.7	75.0 ± 1.7	92.4 ± 1.7	80.3 ± 1.2	83.3 ± 3.4	83.5 ± 1.1	77.5 ± 4.5	90.0 ± 3.7	96.2 ± 1.2
10	95.4 ± 2.8	77.8 ± 1.6	94.3 ± 2.3	87.7 ± 1.0	90.5 ± 3.0	87.9 ± 1.8	79.9 ± 2.7	94.3 ± 4.2	99.5 ± 0.3
11	96.8 ± 0.6	76.1 ± 1.2	86.7 ± 1.3	80.0 ± 1.2	87.4 ± 1.7	85.3 ± 1.7	77.3 ± 0.9	97.1 ± 1.5	99.9 ± 0.1
12	86.4 ± 3.6	72.1 ± 4.2	89.9 ± 4.4	78.6 ± 2.6	84.3 ± 4.3	76.4 ± 4.6	74.8 ± 2.7	92.5 ± 2.7	96.2 ± 0.7
13	73.1 ± 8.3	$33.1 \pm 6.$	97.3 ± 1.2	44.0 ± 0.9	35.3 ± 10.3	39.7 ± 8.4	27.5 ± 2.0	93.2 ± 5.1	97.6 ± 1.0
14	99.0 ± 0.1	97.0 ± 1.4	97.1 ± 0.4	98.6 ± 0.0	98.1 ± 1.2	97.2 ± 2.7	95.9 ± 0.3	96.3 ± 2.6	99.7 ± 0.4
15	99.5 ± 0.3	94.5 ± 2.3	97.8 ± 0.1	99.3 ± 0.4	98.8 ± 0.2	98.6 ± 0.5	96.4 ± 1.6	96.3 ± 4.3	98.0 ± 1.2
OA(%)	92.1 ± 0.7	82.2 ± 0.2	93.8 ± 0.7	87.9 ± 0.1	89.7 ± 0.6	88.6 ± 0.3	83.3 ± 0.5	94.1 ± 1.3	96.6 ± 0.3
AA(%)	92.1 ± 1.2	82.0 ± 0.2	92.7 ± 1.7	87.6 ± 0.1	88.6 ± 0.7	87.6 ± 0.4	82.3 ± 0.7	94.1 ± 1.4	96.7 ± 0.2
Kappa(%)	91.5 ± 0.8	80.8 ± 0.3	93.4 ± 0.9	86.9 ± 0.2	88.9 ± 0.7	87.7 ± 0.3	82.0 ± 0.6	93.7 ± 1.4	96.4 ± 0.3

The visual classification result maps of all the nine algorithms and the ground truth are listed in Fig. 7. For this dataset, it is difficult to analyze the performance of the algorithm using only labeled samples because the distribution of labeled samples is extremely scattered. Therefore, we use all the samples to make the prediction to show full-image classification results. As exhibited in Fig. 7, there are a lot of scattered noisy points, but it is better for edge prediction in RBF-SVM, DDCNN, DCS, mRMR, BS-Nets, and DRLBS. Compared with these methods, MR-Selection can maintain regional consistency well and make the visualization results smoother. MR-Selection considers the spatial contextual correlation between adjacent pixels, which can fully obtain the comprehensive spatial–spectral information. Furthermore, MR-Selection adds an intraclass constraint regularization to force the feature space of samples from the same class to be narrowed so that the features of each class are compact and discriminative. In addition, MR-Selection can select the band combination by using the classification performance as the reward and search it from a huge search space by using the powerful searching ability of RL. It is obvious that MR-Selection maintains a better boundary location in the highway class compared to SICNN and TWCNN.

D. Analysis of Running Time on the Same Dataset

To check the efficiency of the MR-Selection algorithm, Table VII exhibits the training and testing time of all algorithms on three HSI datasets. As shown in Table VII, RBF-SVM spends the least training time because it uses the full bands for training and lacks the stage of band selection. Except for RBF-SVM, the training time of mRMR is the least because of the high efficiency of incremental search in mRMR. Compared with mRMR, DRLBS spends a lot of time because it uses the RL method to find suitable band combinations by constantly interacting with the environment. Unfortunately, the training time of DCS is much longer than that of other algorithms because it needs to train a large number of base classifiers for various candidate band subsets. Among five deep learning-based band selection methods, DDCNN is the fastest because a pretrained 1-D CNN is used to evaluate the band subset without retraining. However, SICNN needs to retrain a 2-D CNN for different candidate band subsets, so it spends more training time. In addition, BS-Nets needs to reconstruct the HSI data, which takes much time to optimize the tricky reconstruction loss. Compared with SICNN and BS-Nets, TWCNN saves the training time by embedding the band selection task into the classification task. It is worth noting that time consumption of MR-Selection has no correlation with the number of training samples. This is because MR-Selection needs to reselect a fixed batch size of samples from the training

TABLE VII

RUNNING TIME OF RBF-SVM, DRLBS, DDCNN, MRMR, SICNN, DCS, BS-NETS, TWCNN, AND MR-SELECTION

Dataset	Method	Training Time(s)	Test Time(s)
Indian Pines	RBF-SVM	0.4 ± 0.1	0.8 ± 0.1
	DRLBS	370.2 ± 1.2	0.3 ± 0.1
	DDCNN	26.3 ± 3.7	1.4 ± 0.3
	mRMR	1.2 ± 0.1	0.3 ± 0.1
	SICNN	1037.2 ± 1.6	1.7 ± 0.2
	DCS	11810.7 ± 125.0	1.8 ± 0.1
	BS-Nets	881.4 ± 1.3	1.8 ± 0.1
	TWCNN	330.9 ± 9.1	1.4 ± 0.0
	MR-Selection	1166.5 ± 9.6	1.9 ± 0.1
Pavia University	RBF-SVM	0.5 ± 0.1	1.2 ± 0.1
	DRLBS	193.2 ± 2.1	0.9 ± 0.1
	DDCNN	36.2 ± 4.1	1.5 ± 0.1
	mRMR	1.9 ± 0.1	0.8 ± 0.1
	SICNN	1586.4 ± 5.4	2.1 ± 0.2
	DCS	16902.5 ± 56.0	1.3 ± 0.2
	BS-Nets	1723.6 ± 4.7	4.9 ± 0.4
	TWCNN	570.5 ± 1.1	3.0 ± 0.0
	MR-Selection	853.7 ± 10.4	4.8 ± 0.0
University of Houston	RBF-SVM	0.6 ± 0.1	1.3 ± 0.1
	DRLBS	227.2 ± 1.6	0.3 ± 0.1
	DDCNN	30.9 ± 2.7	1.2 ± 0.2
	mRMR	1.5 ± 0.1	0.9 ± 0.1
	SICNN	1206.2 ± 3.4	2.7 ± 0.2
	DCS	17161.4 ± 150.5	1.3 ± 0.2
	BS-Nets	1089.6 ± 2.0	1.5 ± 0.2
	TWCNN	460.6 ± 2.2	1.7 ± 0.0
	MR-Selection	1020 ± 5.4	2.2 ± 0.0

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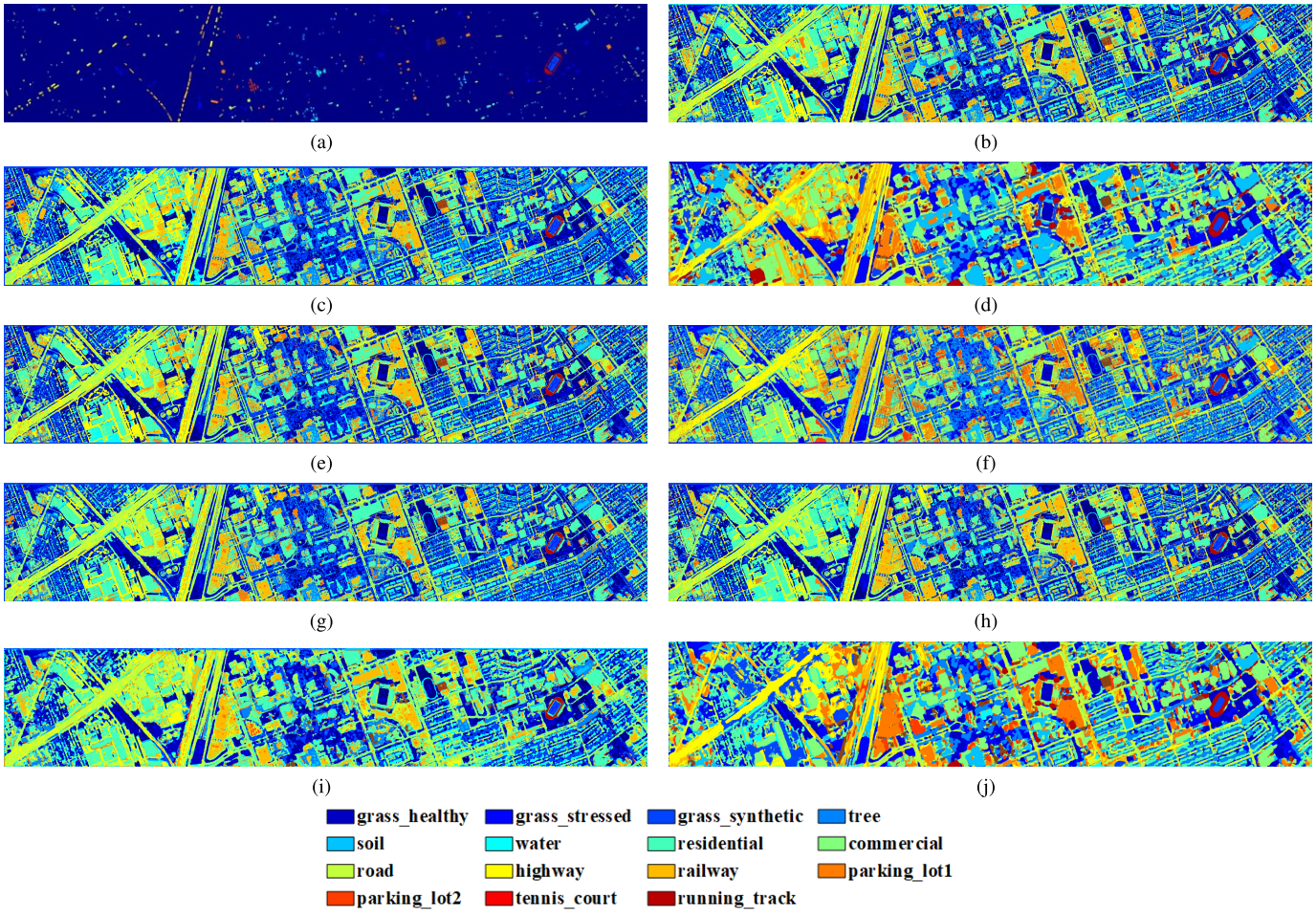


Fig. 7. (a) Ground truth and (b)–(j) visual classification map of the University of Houston dataset by RBF-SVM, mRMR, SICNN, DDCNN, DRLBS, DCS, BS-Nets, TWCNN, and MR-Selection.

set to construct band-based graph in each iteration. Compared with SICNN, BS-Nets, and DCS, MR-Selection saves the training time on the Pavia University and the University of Houston datasets with more training samples. For the test time, all the algorithms can complete the test in seconds. However, it is found that the testing time of the methods based on deep learning is a little longer.

E. Comparison With Other State-of-the-Art Unsupervised Band Selection Methods on the Same Dataset

In this section, MR-Selection is compared to four state-of-art unsupervised band selection methods. These four representative state-of-art unsupervised band selection methods are the gumbel-softmax (GS) trick enabled concrete autoencoder-based method (CAE-UBS) [55], the adaptive distance-based band hierarchy (ADBH) method [56], the hyperbolic clustering-based band hierarchy (HCBH) method [57], and the maximum-information-minimum-redundancy and a modified discrete gravitational search (MIMR-DGSA) method [58].

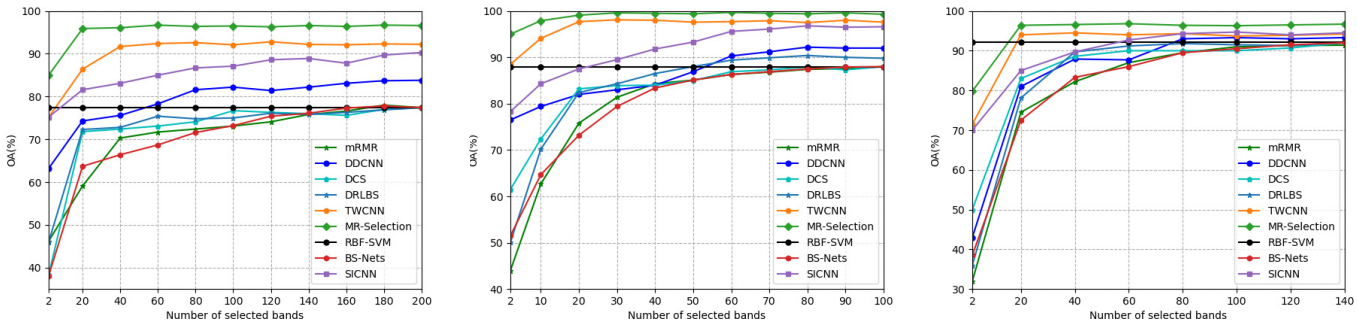
As a deep learning-based unsupervised band selection method, CAE-UBS models the HSI band selection task as a reconstruction task of HSIs by using the autoencoder. In addition, the GS trick is performed to obtain the band subset, which can achieve a discrete autoencoder to select the band subset instead of ranking the learned weight in the process of optimization. In [56], a tree hierarchy-based clustering framework is designed for band selection, where an adaptive distance is proposed to define the distance measurement and avoid the noisy sensitive problem. HCBH [57] is an improved version of ADBH [56]. Specifically, Sun et al. [57] preform the tree hierarchy-based clustering framework in the hyperbolic space and designs adaptive hyperbolic distance to define the distance measurement, which can fully represent the geometric structure of HSI data. In [58], MIMR-DGSA designs a modified discrete gravitational search algorithm to select the candidate band subsets according to the maximum-information-minimum-redundancy criterion.

The classification results on three HSI datasets are listed in Tables VIII. Among these unsupervised algorithms, HCBH shows better classification performance in terms of OA, AA, and Kappa than ADBH and MIMR-DGSA. This is because HCBH can make full use of the geometric structure of HSIs by constructing the tree hierarchy-based clustering in the hyperbolic space. CAE-UBS obtains superior classification performance than MIMR-DGSA. The reason may be that CAE-UBS can learn the autoencoder with a discrete layer directly and avoid the loss of complementary information in the process of sorting well-trained weights. It is worth noting

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TABLE VIII
CLASSIFICATION RESULTS OF CAE-UBS, ADBH, HCBH, MIMR-DGSA, AND MR-SELECTION ON THREE DATASETS

Dataset	Method	OA(%)	AA(%)	Kappa(%)
Indian Pines	CAE-UBS	78.3 ± 1.0	71.1 ± 1.9	75.2 ± 1.1
	ADBH	76.4 ± 0.6	69.1 ± 1.2	73.0 ± 0.7
	HCBH	76.6 ± 0.8	69.4 ± 1.2	73.1 ± 0.9
	MIMR-DGSA	72.2 ± 0.9	61.1 ± 2.2	68.1 ± 0.9
	MR-Selection	96.5 ± 0.1	91.8 ± 1.3	96.0 ± 0.1
Pavia University	CAE-UBS	89.8 ± 0.4	85.7 ± 0.9	86.3 ± 0.5
	ADBH	90.3 ± 0.2	87.3 ± 0.5	87.1 ± 0.3
	HCBH	91.7 ± 0.3	88.8 ± 0.7	88.9 ± 0.4
	MIMR-DGSA	88.1 ± 0.4	83.5 ± 0.6	84.2 ± 0.5
	MR-Selection	99.5 ± 0.1	98.8 ± 0.2	99.3 ± 0.1
University of Houston	CAE-UBS	89.4 ± 1.1	88.7 ± 0.8	88.5 ± 1.2
	ADBH	88.5 ± 0.2	88.0 ± 0.3	87.5 ± 0.2
	HCBH	90.0 ± 0.4	89.0 ± 0.4	89.2 ± 0.4
	MIMR-DGSA	88.6 ± 0.8	87.5 ± 0.9	87.6 ± 0.9
	MR-Selection	96.6 ± 0.3	96.7 ± 0.2	96.4 ± 0.3



(a) (b) (c)

Fig. 8. OA curves of RBF-SVM, mRMR DRLBS, BS-Nets, DCS, TWCNN, SICNN, DDCNN, and MR-Selection as the number of selected bands increases from 2 to 200, 2 to 100, and 2 to 140 on (a) Indian Pines, (b) Pavia University, and (c) University of Houston datasets.

that MR-Selection achieves the best classification performance in terms of OA, AA, and Kappa on the three HSI datasets.

F. Sensitivity to the Number of Selected Bands on the Same Dataset

In this section, the classification performance of all algorithms with different numbers of selected bands on the same HSI datasets is analyzed. Fig. 8 shows the variation curve of the OA value of different band selection methods with different numbers of selected bands on three HSI datasets. The range of the number of selected bands is set to [2, 200] and [2, 140] with intervals of 20 spectral bands on the Indian Pines and University of Houston datasets. The range of the number of selected bands is set to [2, 100] with intervals of ten spectral bands on the Pavia University. Since RBF-SVM uses all the spectral bands for classification, its OA value remains unchanged in the entire range. As shown in Fig. 8, mRMR, DRLBS, BS-Nets, and DCS have sharp changes within the range of [2, 20] on three HSI datasets. In this case, too few bands selected by these methods may not have enough discriminative features for the classifiers to distinguish the samples of different classes. However, SICNN, TWCNN, and MR-Selection are relatively slow. This may be that, in these methods, on contrary, the OA value of SICNN continues to improve with the increase of the number of selected bands due to the addition of more discriminative bands. Compared to TWCNN and SICNN, the classification performance of MR-Selection is optimal in all the cases, which further proves that MR-selection tends to select more discriminative bands.

G. Classification Results on the Unseen Dataset

To confirm the performance of MR-Selection on unseen datasets, the training and inference are implemented on different datasets. As far as we know, there is currently no work on zero-shot band selection of HSIs. We use some intuitive operations to make the comparison algorithms work on zero-shot band selection of HSIs. Specifically, among all the training datasets, first, find the dataset with the largest number of bands. Then, fill 0 to other datasets to maintain the same number of spectral bands with it. For mRMR, calculate the score of the candidate band subset of each dataset, and take the average score of all the datasets as the final score of the candidate band subset. In the stage of evaluation, the evaluation scores of all the datasets are averaged as the final score to update the candidate band combination. SICNN is the same as that of DDCNN except that 2-D CNN is retrained for every candidate band combination corresponding to each dataset during the training process. For DCS, we average the selection frequency of the base classifier corresponding to each dataset as the final selection of candidate band combination. In TWCNN, we set up multiple classifiers at the end of the network to update the parameter of the network together. For BS-Nets, multiple construction networks are set to obtain the weight value of each band in different datasets and, finally, select the band combination based on the average weight value.

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TABLE IX

ZERO-SHOT BAND SELECTION CLASSIFICATION RESULTS OF MRMR, SICNN, DDCNN, DRLBS, DCS, BS-NETS, TWCNN, AND MR-SELECTION ON THE INDIAN PINES DATASET

	mRMR	SICNN	DDCNN	DRLBS	DCS	BS-Nets	TWCNN	MR-Selection
OA(%)	70.8 ± 0.8	85.0 ± 3.2	75.7 ± 1.7	72.6 ± 1.2	75.7 ± 1.0	74.0 ± 0.3	90.5 ± 1.3	95.9 ± 0.6
AA(%)	60.9 ± 1.3	78.6 ± 3.4	71.6 ± 1.9	60.0 ± 1.5	66.9 ± 3.6	64.3 ± 1.6	$78.6v2.5$	89.2 ± 1.7
Kappa(%)	66.6 ± 0.9	82.8 ± 3.7	72.2 ± 1.8	68.5 ± 1.4	72.0 ± 1.2	70.0 ± 0.3	89.1 ± 1.5	95.4 ± 0.7

TABLE X

ZERO-SHOT BAND SELECTION CLASSIFICATION RESULTS OF MRMR, SICNN, DDCNN, DRLBS, DCS, BS-NETS, TWCNN, AND MR-SELECTION ON THE PAVIA UNIVERSITY DATASET

	mRMR	SICNN	DDCNN	DRLBS	DCS	BS-Nets	TWCNN	MR-Selection
OA(%)	73.4 ± 0.4	94.5 ± 0.6	83.8 ± 1.0	86.2 ± 0.7	82.8 ± 0.5	79.5 ± 0.4	94.2 ± 1.6	99.4 ± 0.1
AA(%)	63.2 ± 1.0	90.4 ± 1.6	79.7 ± 0.9	79.5 ± 2.9	69.3 ± 0.8	66.7 ± 1.1	89.8 ± 3.6	98.5 ± 0.1
Kappa(%)	63.3 ± 0.5	92.7 ± 0.9	78.4 ± 1.3	81.2 ± 1.0	76.4 ± 0.8	71.2 ± 0.6	92.3 ± 2.1	99.2 ± 0.1

TABLE XI

ZERO-SHOT BAND SELECTION CLASSIFICATION RESULTS OF MRMR, SICNN, DDCNN, DRLBS, DCS, BS-NETS, TWCNN, AND MR-SELECTION ON THE HOUSTON UNIVERSITY DATASET

	mRMR	SICNN	DDCNN	DRLBS	DCS	BS-Nets	TWCNN	MR-Selection
OA(%)	80.6 ± 1.0	88.9 ± 1.7	86.0 ± 0.1	79.9 ± 0.8	76.8 ± 0.8	79.7 ± 0.5	91.4 ± 2.5	96.1 ± 0.1
AA(%)	79.5 ± 1.1	87.3 ± 1.9	85.2 ± 0.4	78.9 ± 0.9	76.2 ± 0.6	79.2 ± 0.4	90.8 ± 2.3	95.8 ± 0.3
Kappa(%)	79.0 ± 1.1	88.0 ± 1.8	84.9 ± 0.2	78.2 ± 0.8	74.9 ± 0.8	78.1 ± 0.5	90.6 ± 2.7	95.7 ± 0.1

In the experiment, six HSI datasets are used. Among them, Salinas, KSC, and Washington datasets are used for training, and the remaining Indian Pines, Pavia University, and University of Houston datasets are used for testing. Tables IX–XI record the average values and corresponding standard deviations of OA, AA, and Kappa indexes after each algorithm is run on these three datasets for 30 times independently. The optimal OA, AA, and Kappa indexes are highlighted in the gray region.

As shown in Tables IX–XI, MR-Selection keeps the best performance on the OA, AA, and Kappa indexes, which proves that MR-Selection can select suitable bands for classification on unseen datasets, while other comparison algorithms have poor performance on some datasets. This is mainly because MR-Selection can learn the metaknowledge among different band selection tasks during the training process, and the metaknowledge can be used to select the bands reasonably during the test phase. In contrast, other comparison algorithms provide extremely limited guidance for band selection on the new dataset. In addition, the comparison algorithms are biased in zero-shot band selection. It is possible that the selected bands have better results for a certain dataset but performs poorly for other datasets. However, the ability of MR-Selection has outstanding generalization ability on any new dataset by extracting the correlation among different band selection tasks.

H. Sensitivity to the Number of Selected Bands on the Unseen Dataset

In this section, the classification performance of all algorithms with different numbers of selected bands on cross HSI datasets is analyzed. Fig. 9 shows the change trend of the OA of all the algorithms on three datasets. For RBF-SVM, we use the full bands for training, so the trend of its OA curve does not change in the whole variation range. For all the algorithms except RBF-SVM, the number of bands that needs to be selected is analyzed in a variation range of [2, 200] on the Indian Pines dataset. For Pavia University and University of Houston datasets, the number of bands that need to be selected is in the variation range of [2, 100] and [2, 140], respectively. In addition, the OA value of every 20 bands is recorded within the range of variation.

As shown in Fig. 9, the OA values of the mRMR, DDCNN, DCS, DRLBS, and BS-Nets algorithms have a sharp increase in the range of [2, 20] on all the three datasets. The promotion of TWCNN, SICNN, and MR-Selection is relatively slow, especially for MR-Selection. However, the OA value of other algorithms has improved in varying degrees with the increasing number of spectral bands when more than 20 spectral bands are selected. The reason is that the discriminative bands are newly selected. For example, the OA value of most comparison algorithms on the Indian Pines dataset is stable when 100 bands are selected, while MR-Selection is stable when 20 bands are selected. Similar experimental results are obtained on the Pavia University and Houston datasets. It proves that MR-Selection can select the discriminative bands even when the number of bands is relatively small. In addition, MR-Selection completely exceeds other algorithms when any number of bands are selected on the three datasets, which further proves the effectiveness of the MR-Selection algorithm.

I. Analysis With Selected Spectral Bands on the Unseen Dataset

In this section, 30 spectral bands on three datasets are selected to evaluate the selected bands more intuitively for all the algorithms except RBF-SVM. As shown in Fig. 10, the upper part of the corresponding subgraph of each dataset shows the positions of 30 selected bands, and each point on the line represents the corresponding position of each selected band within the entire band range. The lower of Fig. 10(a)–(c) shows the Shannon entropy of each band, which is used to

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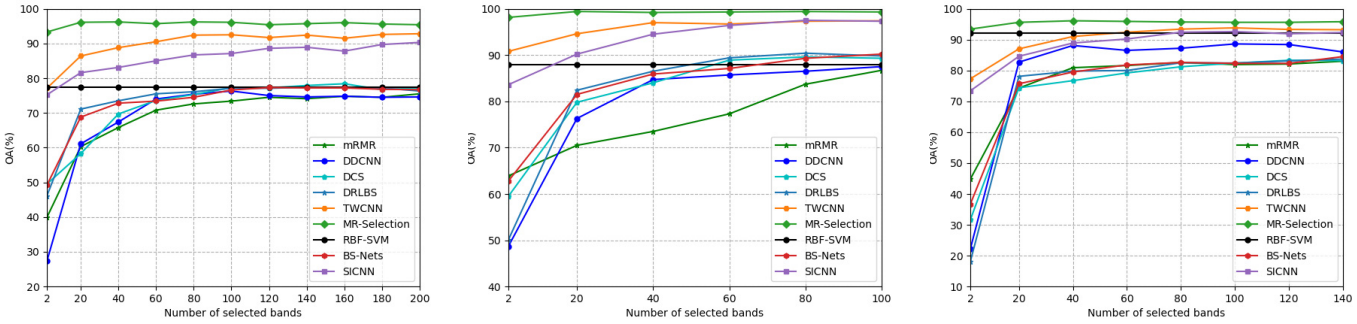


Fig. 9. OA curves of RBF-SVM, mRMR DRLBS, BS-Nets, DCS, TWCNN, SICNN, DDCNN, and MR-Selection as the number of selected bands increases from 2 to 200, 2 to 100, and 2 to 140 on (a) Indian Pines, (b) Pavia University, and (c) University of Houston dataset, respectively.

TABLE XII
CLASSIFICATION PERFORMANCE IN ABLATION EXPERIMENTS

Dataset	Conditions	OA(%)	AA(%)	Kappa(%)
Indian Pines	MR-Selection without spectral correlation	91.0 ± 2.5	81.9 ± 3.8	89.8 ± 2.9
	MR-Selection without spatial correlation	95.5 ± 1.4	89.0 ± 2.5	94.9 ± 1.6
	MR-Selection without two-stage optimization	94.5 ± 0.2	87.7 ± 0.3	93.8 ± 0.2
	MR-Selection	95.9 ± 0.6	89.2 ± 1.7	95.4 ± 0.7
Pavia University	MR-Selection without spectral correlation	94.6 ± 2.4	92.3 ± 3.2	92.9 ± 3.1
	MR-Selection without spatial correlation	98.9 ± 0.2	98.2 ± 0.4	98.6 ± 0.3
	MR-Selection without two-stage optimization	97.7 ± 0.1	95.7 ± 0.1	96.9 ± 0.2
	MR-Selection	99.4 ± 0.1	98.5 ± 0.1	99.2 ± 0.1
University of Houston	MR-Selection without spectral correlation	91.9 ± 0.7	91.7 ± 0.7	91.2 ± 0.7
	MR-Selection without spatial correlation	95.4 ± 0.2	94.9 ± 0.1	95.0 ± 0.2
	MR-Selection without two-stage optimization	93.4 ± 0.2	93.1 ± 0.2	92.8 ± 0.2
	MR-Selection	96.1 ± 0.1	95.8 ± 0.3	95.7 ± 0.1

represent the information of each band in HSI data. The comparison algorithms themselves are not designed for the unseen dataset. To realize zero-shot band selection, we have to implement a zero-filling strategy for these methods. These methods choose some bands that exceed the original number of bands on the Pavia University dataset and the University of Houston datasets. In HSIs, there is highly correlated between adjacent bands. It is easy for these adjacent bands to lead to information redundancy. The bands selected by mRMR, DCS BS-Nets, and DDCNN are adjacent and denser. Compared with these methods, MR-Selection, DRLBS, TWCNN, and SICNN select relatively wider and scattered bands, which have more chance to provide complementary information. However, TWCNN and SICNN choose some bands with low information entropy, which may easily lead to poor classification performance. Compared with these methods, MR-Selection selects more discriminative and complementary band subsets. For all the comparison algorithms, only the results of band selection are learned from the training datasets, which cannot select the band combination by utilizing the unique characteristics of each unseen test dataset. Surprisingly, MR-Selection can select the appropriate band combination adaptively by utilizing the metaknowledge from the training datasets and the unique information of each test dataset. It has a better generalization ability in unseen datasets.

J. Analysis of Hyperparameter in MR-Selection on Unseen Dataset

To analyze the impact of the penalty factor on the band combination selected by MR-Selection, different values are investigated to validate the classification performance with fixed other parameters. The penalty factor α in the reward is used to punish the repeated selection of bands. It is set in the interval of $[-0.00001, -1]$. In three HSI datasets, the number of spectral bands that need to be selected is fixed as three. This is because, when the number of selected bands is small, the performance change of the proposed method is more obvious for the penalty factors with different values. At this time, it is clearer to display the effect of the penalty factor on the proposed method. Fig. 11 shows the corresponding OA and standard deviations of MR-Selection with different α values on three datasets. As shown in Fig. 11, if the penalty factor α is too small, it is easy to make MR-Selection fall into choosing repetitive bands. In this case, the agent may be unable to explore more bands. On the contrary, if α is too large, it is difficult for the agent to reach the convergence. Therefore, whether this parameter is too small or too large, it will lead to performance degradation on three HSI datasets. Thus, -0.0001 is the most appropriate penalty factor.

K. Ablation Experiments in MR-Selection on Unseen Dataset

In this section, the ablation experiments on three HSI datasets are performed to demonstrate the effectiveness of spectral correlation in the graph, spatial correlation in the graph, and two-stage optimization in the proposed method. To demonstrate the contribution of each module, the OA, AA, and Kappa results of MR-Selection without different modules and MR-Selection are shown in Table XII.

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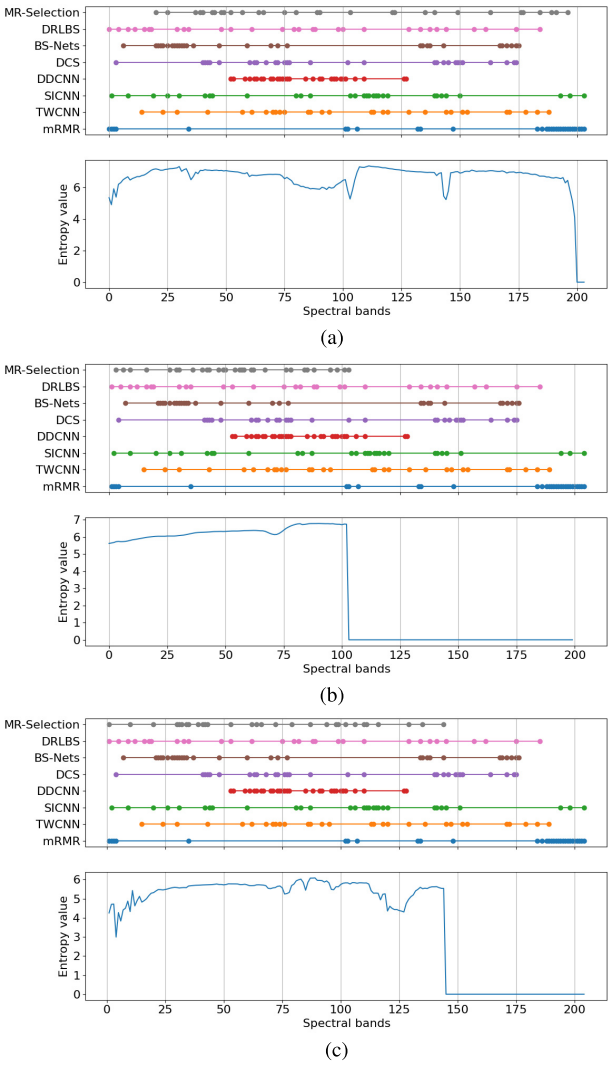


Fig. 10. 30 bands selected by different band selection methods. The above subgraph represents the result of 30 selected bands, and the below subgraph represents the entropy value of all bands. (a) Indian Pines, (b) Pavia University, and (c) University of Houston datasets.

As displayed in Table XII, when the spectral correlation in the graph is removed and spatial correlation is retained, the classification performance of MR-Selection degrades most severely. This shows that the spectral correlation is crucial in MR-Selection. The spectral correlation makes the band-based graph have more discriminative information by calculating the correlation between the graph nodes. The spatial correlation is to calculate the distance between spectral bands in physical space, which can only be a supplement to the discriminative information. If the two-stage optimization method is removed, the classification performance of MR-Selection also decreases. When there is no two-stage optimization, the agent is affected by different optimization directions from multiple tasks, which makes it difficult to find the optimization direction suitable for all the tasks. In a word, the combination of spectral correlation in the graph, spatial correlation in the graph, and two-stage optimization enables MR-Selection to achieve outstanding band selection and classification performance. Missing any one of them will lead to the degradation of classification performance.

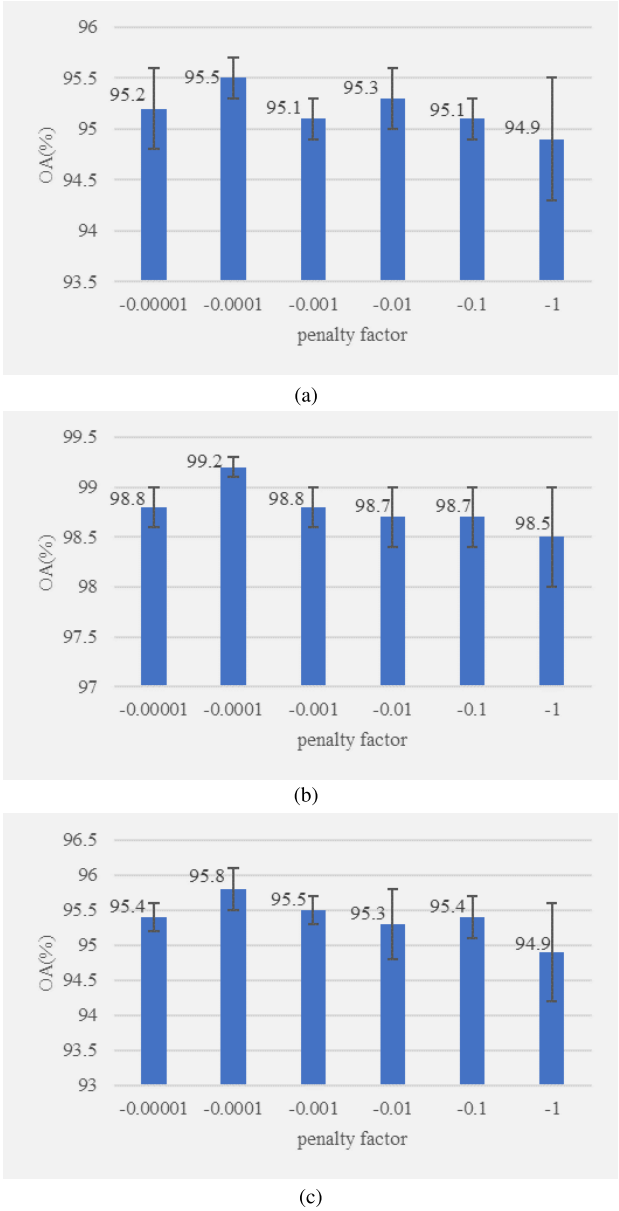


Fig. 11. OA value of MR-Selection with different values of the penalty factor on (a) Indian Pines, (b) Pavia University, and (c) University of Houston dataset, respectively. The penalty factor is set in the interval of $[-0.00001, -1]$.

V. Conclusion

At present, most of the deep learning techniques are data-driven, and the performance is outstanding under the same distribution of test and training samples. However, the generalization ability is a bottleneck problem that restricts the development of deep learning. The unseen dataset processing is a concrete instance to verify the generalization ability. In this article, a novel zero-shot band selection method, MR-Selection, is designed. Band selection tasks on different seen datasets are regarded as the subtask in the metalearning, and the metatask is to extract the inherent correlation between different band selection tasks. To solve the combinatorial optimization problem of band selection from the huge searching space, A2C-RL is designed. In A2C-RL, dynamic structure-aware GCN is constructed in the policy and critic networks, which can alleviate the problem of representation

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inconsistency of different band selection tasks. Furthermore, a two-stage optimization is implemented for the coordination of diversity objectives from various tasks in A2C-RL. When the agent is trained, a generalized policy network for band selection is obtained, which can guide the agent to select the bands effectively on unseen datasets.

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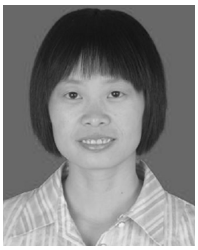
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